ETSI EN 300 328 V2.2.2 (2019-07)

TEST REPORT

For

Quanzhou STAYTRUE Electronic Technology Co., Ltd

Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China

Tested Model: SDC-LE-02HS

Report Type:	**Product Name:**
Original Report	Module SDC-LE-02HS

Report Number:	2607T30160E-RF-02
Report Date:	2026-06-26
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REPORT REVISION HISTORY

Number of Revisions	Report No.	Version	Issue Date	Description
0	2607T30160E-RF-02	R1V1	2026-06-26	Initial Release

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Applicant:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Applicant Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Manufacturer:	Quanzhou STAYTRUE Electronic Technology Co., Ltd
Manufacturer Address:	Unit 1A-2, 1st Floor, Building 1, Fengshu Logistics Park, No. 920, Binhu South Road, Taiwan Investment Zone, Quanzhou, Fujian Province, China
Product Name:	Module SDC-LE-02HS
Tested Model:	SDC-LE-02HS
Multiple Model(s):	N/A
Trade Mark:	SDC-LE-02HS
Power Supply:	DC 3.3V
RF Function:	Classic BT
Maximun EIRP:	3.55dBm
Operating Band/Frequency:	2402-2480 MHz
Channel Number:	79
Channel Separation:	1 MHz
Modulation Type:	GFSK, π/4-DQPSK, 8DPSK
★Maximum Antenna Gain:	1.39 dBi
EUT Received Status:	Good
Note: *1. The maximum antenna gain is provided by the applicant.* *2. All measurement and test data in this report was gathered from production sample serial number: 3LJC-1(RF conducted), 3LJC-2(Radiated Spurious Emission) (Assigned by BACL (Fujian). The EUT was received on 2026-05-08).*	

Objective

This report is prepared for *Quanzhou STAYTRUE Electronic Technology Co., Ltd* in accordance with ETSI EN 300 328 V2.2.2 (2019-07), Wideband transmission systems; Data transmission equipment operating in the 2,4 GHz band; Harmonised Standard for access to radio spectrum.

The objective is to determine the compliance of EUT with ETSI EN 300 328 V2.2.2 (2019-07).

Test Methodology

All measurements contained in this report were conducted with ETSI EN 300 328 V2.2.2 (2019-07).

Measurement Uncertainty

Item	Frequency Range	U_{cispr}	U_{lab}= 2 U_c (y) （Confidence of 95%）
Radiated Spurious Emission	30MHz~1GHz	5.06dB	4.68dB
	1GHz~18GHz	5.12dB	5.00dB
Transmitter Conducted Power			±0.43 dB
Occupy Bandwidth			1.69%
Receiver Blocking characteristics			3.2dB
Voltage (DC)			±0.4%
Temperature			±1℃
Humidity			±5%

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Fujian) to collect test data is located on the Unit 302, No.902, Meifeng South Road, Tong'an District, Xiamen, Fujian, China.

SYSTEM TEST CONFIGURATION

Test Mode and Voltage

The system was configured for testing in a typical mode (as normally used by a typical user).	
Test mode:	Test Mode 1: Transmitting Test Mode 2: Receiving
Test voltage:	DC 3.3V
Remark:	During all emission tests, the EUT was configured to measure its highest possible emission level and the worst case's test data was presented in this test report.

Description of Test Configuration

Channel list:

Channel	Frequency (MHz)	Channel	Frequency (MHz)
0	2402	40	2442
1	2403	…	…
2	2404	…	…
…	…	…	…
…	…	78	2480
39	2441	/	/

EUT was tested with Channel 0, 39 and 78.

★The extreme temperature test conditions which were declared by the manufacturer and the normal conditions are as below:

NT: Normal Temperature +25℃
LT: Low Temperature -40℃
HT: High Temperature +85℃

★EUT Exercise Software

RF test tool: BT_Tool.exe

Test Modes	Power Level Setting		
	Lowest Channel	Middle Channel	Highest Channel
GFSK	6	6	6
π/4-DQPSK	6	6	6
8DPSK	6	6	6

Note: The power level setting was declared by the applicant.

Special Accessories

No special accessory.

Equipment Modificationss

No modification was made to the EUT.

Support Equipment List and Details

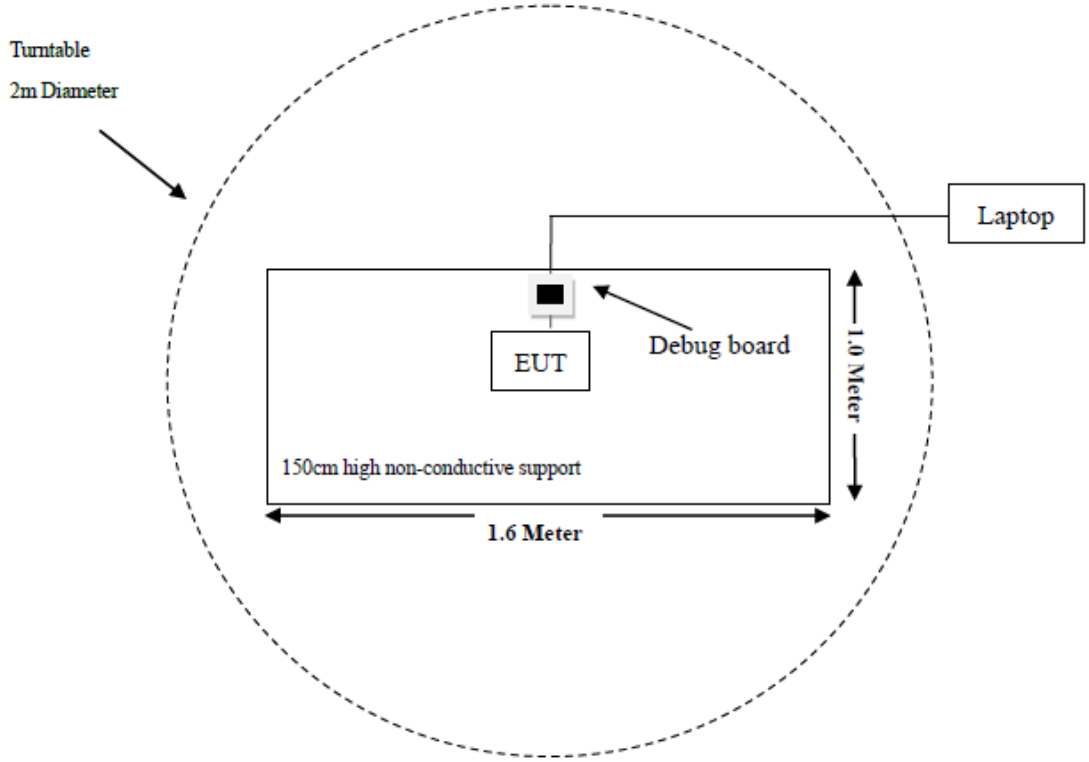
Manufacturer	Description	Model	Serial Number
Lenovo	Laptop	T480	PF1P5K4F
Unknown	Debug board	Unknown	Unknown

External Cable

Cable Description	Length(m)	From Port	To Port
Jumper Wire	0.2	EUT	Debug board
USB cable	10	Debug board	Laptop

Block Diagram of Test Setup

Radiated Spurious Emission (30MHz-12.75GHz):



SUMMARY OF TEST RESULTS

ETSI EN 300 328 V2.2.2 (2019-07)	Description of Test	Test Result
Clause 4.3.1.2	RF Output Power	Compliant
Clause 4.3.1.3	Duty Cycle, Tx-sequence, Tx-gap	Not Applicable (See Note 1)
Clause 4.3.1.4	Accumulated Transmit Time, Frequency Occupation and Hopping Sequence	Compliant
Clause 4.3.1.5	Hopping Frequency Separation	Compliant
Clause 4.3.1.6	Medium Utilization (MU) factor	Not Applicable (See Note 1)
Clause 4.3.1.7	Adaptivity (Adaptive Frequency Hopping)	Not Applicable (See Note 2)
Clause 4.3.1.8	Occupied Channel Bandwidth	Compliant
Clause 4.3.1.9	Transmitter Unwanted Emissions in the out-of-Band Domain	Compliant
Clause 4.3.1.10	Transmitter Unwanted Emissions in the Spurious Domain	Compliant
Clause 4.3.1.11	Receiver Spurious Emissions	Compliant
Clause 4.3.1.12	Receiver Blocking	Compliant
Clause 4.3.1.13	Geo-location Capability	Not Applicable (See Note 3)

Note:
1. The EUT is adaptive equipment, this item only for non-adaptive equipment.
2. The requirement does not apply for equipment with a maximum declared RF Output power of less than 10 dBm e.i.r.p.
3. The supplier declared that the equipment is unable to perform this function.

TEST EQUIPMENT LIST

Test Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due Date
Radiated Spurious Emissions Below 1 GHz					
EMI Test Receiver	Rohde & Schwarz	ESR	103103	2026/01/21	2027/01/20
Antenna	Sunol Sciences	JB6	A122022-5	2023/07/27	2026/07/26
Amplifier	Sonoma	310B	120903	2026/01/22	2027/01/21
Coaxial Cable	XINHANGWEIBO	XH400T-N-4M	CC002	2026/01/22	2027/01/21
Coaxial Cable	XINHANGWEIBO	XH460B-N-2M	CC006	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH460B-N-12M	CC007	2026/01/20	2027/01/19
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20
Dipole antenna	EMCO	3121C	9209-860	N/A	N/A
Radiated Spurious Emissions Above 1 GHz					
Spectrum Analyzer	Rohde & Schwarz	FSV40-N	102051	2026/01/21	2027/01/20
Filter Switch Unit	Decentest	DT7220FSU	DS79904	2026/01/20	2027/01/19
Multiplex Switch Test Control Set	Decentest	DT7220SCU	DS79901	2026/01/21	2027/01/20
Horn Antenna	EMCO	3115	9002-3355	2024/11/19	2027/11/18
Double Ridge Guide Horn Antenna	A.H.Systems	SAS-571	1980	2023/07/28	2026/07/27
Preamplifier	GLOBAL	1313-A100M18G	4121301	2026/01/14	2027/01/13
Coaxial Cable	XINHANGWEIBO	XH800A-N-6M	CC003	2026/01/20	2027/01/19
Coaxial Cable	XINHANGWEIBO	XH800A-N-1M	CC005	2026/01/20	2027/01/19
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20

Test Equipment	Manufacturer	Model	Serial Number	Calibration Date	Calibration Due Date
RF Conducted Test					
Spectrum Analyzer	Rohde & Schwarz	FSU 26	100405	2026/01/21	2027/01/20
constant temperature and humidity testing machine	BACL	BTH-150-60	30234	2025/07/28	2026/07/27
Microwave Analog Signal Generator	Agilent	N5183A	MY47420335	2026/01/21	2027/01/20
Wideband Radio Communication Tester	Rohde & Schwarz	CMW500	122920	2026/01/22	2027/01/21
USB Wideband Power Sensor	Boonton	55318	8934	2025/09/08	2026/09/07
2 Way power divider	narda	4426LB-2	1661	Each time	N/A
Attenuator	Agilent	8496B	MY42140574	Each time	N/A
Attenuator	HP	8494B	2550A12096	Each time	N/A
Directional coupler	narda	4226-20	08412	2026/01/22	2027/01/21
Coaxial Cable	YINSAIGE	SS405-0.5M	CC018	2025/09/25	2026/09/24
Coaxial Cable	YINSAIGE	SS405-0.5M	CC019	2025/09/25	2026/09/24

Statement of Traceability: Bay Area Compliance Laboratories Corp. (Fujian) attests that all calibrations have been performed in accordance to requirements that traceable to National Primary Standards and International System of Units (SI).

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.2 – RF OUTPUT POWER

Applicable Standard

This requirement applies to all types of FHSS equipment.

The RF output power is defined as the mean equivalent isotropically radiated power (e.i.r.p.) of the equipment during atransmission burst.

Limit

The RF output power for FHSS equipment shall be equal to or less than 20 dBm.

For Non-adaptive FHSS equipment, the manufacturer may have declared a reduced RF Output Power (see clause 5.4.1 m)) and associated Duty Cycle (see clause 5.4.1 e)) that will ensure that the equipment meets the requirement for the Medium Utilization (MU) factor further described in clause 4.3.1.6. This is verified by the conformance test referred to in clause 4.3.1.6.4.

For non-adaptive FHSS equipment, where the manufacturer has declared an RF output power lower than 20 dBm e.i.r.p., the RF output power shall be equal to or less than that declared value.

This limit shall apply for any combination of power level and intended antenna assembly.

Test Procedure

The test procedure shall be as follows:

Step 1:

• Use a fast power sensor with a minimum sensitivity of -40 dBm and capable of minimum 1 MS/s.
• Use the following settings:
  - Sample speed 1 MS/s or faster.
  - The samples shall represent the RMS power of the signal.
  - Measurement duration: For non-adaptive equipment: equal to the observation period defined in clause 4.3.1.3.2 or clause 4.3.2.4.2. For adaptive equipment, the measurement duration shall be long enough to ensure a minimum number of bursts (at least 10) is captured.
    For adaptive equipment, to increase the measurement accuracy, a higher number of bursts may be used.

Step 2:

• For conducted measurements on devices with one transmit chain:
  - Connect the power sensor to the transmit port, sample the transmit signal and store the raw data. Use these stored samples in all following steps.
• For conducted measurements on devices with multiple transmit chains:
  - Connect one power sensor to each transmit port for a synchronous measurement on all transmit ports.
  - Trigger the power sensors so that they start sampling at the same time. Make sure the time difference between the samples of all sensors is less than 500 ns.
  - For each individual sampling point (time domain), sum the coincident power samples of all ports and store them. Use these summed samples as the new stored data set.

Step 3:

• Find the start and stop times of each burst in the stored measurement samples.
The start and stop times are defined as the points where the power is at least 30 dB below the highest value of the stored samples in step 2.

NOTE 2: In case of insufficient dynamic range, the value of 30 dB may need to be reduced appropriately.

Step 4:

• Between the start and stop times of each individual burst calculate the RMS power over the burst using the formula below. The start and stop points shall be included.Save these P_{burst} values, as well as the start and stop times for each burst.

$$P_{burst} = \frac{1}{k}\sum_{n=1}^{k} P_{sample}(n)$$

With 'k' being the total number of samples and 'n' the actual sample number

Step 5:

• The highest of all P_{burst} values (value "A" in dBm) will be used for maximum e.i.r.p. calculations.

Step 6:

• Add the (stated) antenna assembly gain "G" in dBi of the individual antenna.
• If applicable, add the additional beamforming gain "Y" in dB.
• If more than one antenna assembly is intended for this power setting, the maximum overall antenna gain (G or G + Y) shall be used.
• The RF Output Power (P) shall be calculated using the formula below:

$$\mathbf{P = A + G + Y}$$

• This value, which shall comply with the limit given in clauses 4.3.1.2.3 or clause 4.3.2.2.3, shall be recorded in the test report.

Test Setup Block diagram



Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

Test Data

Test Mode:	Transmitting			Test Engineer:			Spike Gao
Test Date:	2026-05-29			Environment:			Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa
Mode	Channel	Conducted Output Power (dBm)			Result (dBm)			Limit (dBm)
		LT	NT	HT	LT	NT	HT	
GFSK	Hopping	2.16	2.09	2.02	3.55	3.48	3.41	≤ 20
π/4-DQPSK	Hopping	1.88	1.82	1.76	3.27	3.21	3.15	
8DPSK	Hopping	1.86	1.79	1.73	3.25	3.18	3.12	

Note :
1.The antenna gain is 1.39 dBi.
2.The extreme operating temperature range was declared by the manufacturer.

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.4 – ACCUMULATED TRANSMIT TIME, FREQUENCY OCCUPATION AND HOPPING SEQUENCE

Applicable Standard

These requirements apply to all types of FHSS equipment.

The Accumulated Transmit Time is the total of the transmitter 'on' times, during an observation period, on a particular hopping frequency.

The Frequency Occupation is the number of times that each hopping frequency is occupied within a given period. A hopping frequency is considered to be occupied when the equipment selects that frequency from the hopping sequence.The equipment may be transmitting, receiving or stay idle during the Time spent on that hopping frequency.

The Hopping Sequence of a frequency hopping equipment is the unrepeated pattern of the hopping frequencies used by the equipment.

Limit:

For Adaptive FHSS equipment:

Adaptive FHSS equipment shall be capable of operating over a minimum of 70 % of the band specified in table 1.
The Accumulated Transmit Time on any hopping frequency shall not be greater than 400 ms within any observation period of 400 ms multiplied by the minimum number of hopping frequencies (N) that have to be used..

In order for the FHSS equipment to comply with the Frequency Occupation requirement, it shall meet either of the following two options:

• Option 1: Each hopping frequency of the hopping sequence shall be occupied at least once within a period not exceeding four times the product of the dwell time and the number of hopping frequencies in use.

• Option 2: The occupation probability for each frequency shall be between ((1/U) × 25 %) and 77 % where U is the number of hopping frequencies in use.

The hopping sequence(s) shall contain at least N hopping frequencies at all times, where N is either 15 or the result of 15 MHz divided by the minimum Hopping Frequency Separation in MHz, whichever is the greater.

Test Procedure

The test procedure shall be as follows:

Step 1:

- The output of the transmitter shall be connected to a spectrum analyzer or equivalent.
- The analyzer shall be set as follows:
  - Centre Frequency: Equal to the hopping frequency being investigated
  - Frequency Span: 0 Hz
  - RBW: ~ 50 % of the Occupied Channel Bandwidth
  - VBW: ≥ RBW

- Detector Mode: RMS
- Sweep time: Equal to the applicable observation period (see clause 4.3.1.4.3.1 or clause 4.3.1.4.3.2)
- Number of sweep points: 30 000
- Trace mode: Clear / Write
- Trigger: Free Run

Step 2:

• Save the trace data to a file for further analysis by a computing device using an appropriate software application or program.

Step 3:

• Identify the data points related to the frequency being investigated by applying a threshold.

The data points resulting from transmissions on the hopping frequency being investigated are assumed to have much higher levels compared to data points resulting from transmissions on adjacent hopping frequencies. If a clear determination between these transmissions is not possible, the RBW in step 1 shall be further reduced. In addition, a channel filter may be used.

• Count the number of data points identified as resulting from transmissions on the frequency being investigated and multiply this number by the time difference between two consecutive data points.

Step 4:

• The result in step 3 is the Accumulated Transmit Time which shall comply with the limit provided in clause 4.3.1.4.3.1 or clause 4.3.1.4.3.2 and which shall be recorded in the test report.

Step 5:

NOTE 1: This step is only applicable for equipment implementing Option 1 in clause 4.3.1.4.3.1 or Option 1 in clause 4.3.1.4.3.2 for complying with the Frequency Occupation requirement and the manufacturer decides to demonstrate compliance with this requirement via measurement.

• Make the following changes on the analyser and repeat step 2 and step 3.

Sweep time: 4 ×Dwell Time ×Actual number of hopping frequencies in use.

The hopping frequencies occupied by the equipment without having transmissions during the dwell time (blacklisted frequencies) should be taken into account in the actual number of hopping frequencies in use. If this number cannot be determined (number of blacklisted frequencies unknown) it shall be assumed that the equipment uses the maximum possible number of hopping frequencies.

• The result shall be compared to the limit for the Frequency Occupation defined in clause 4.3.1.4.3.1, Option 1 or clause 4.3.1.4.3.2, Option 1. The result of this comparison shall be recorded in the test report.

Step 6:

• Make the following changes on the analyzer:
- Start Frequency: 2 400 MHz
- Stop Frequency: 2 483,5 MHz
- RBW: ~ 50 % of the Occupied Channel Bandwidth (single hopping frequency)

- VBW: ≥ RBW
- Detector Mode: Peak
- Sweep time: 1 s; this setting may result in long measuring times. To avoid such long measuring times, an FFT analyser may be used
- Trace Mode: Max Hold
- Trigger: Free Run

• Wait for the trace to stabilize. Identify the number of hopping frequencies used by the hopping sequence.

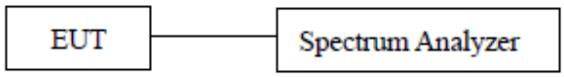
• The result shall be compared to the limit (value N) defined in clause 4.3.1.4.3.1 or clause 4.3.1.4.3.2. This value shall be recorded in the test report.

NOTE 2: For equipment with blacklisted frequencies, it might not be possible to verify the number of hopping frequencies in use. However, they shall comply with the requirement for Accumulated Transmit Time and Frequency Occupation assuming the minimum number of hopping frequencies (N) defined in clause 4.3.1.4.3.1 or clause 4.3.1.4.3.2 is used.

Step 7:

• For adaptive frequency hopping equipment, it shall be verified whether the equipment uses 70 % of the band specified in table 1. This verification can be done using the lowest and highest -20 dB points from the total spectrum envelope obtained in step 6. The result shall be recorded in the test report.

Test Setup Block diagram



Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

Test Data

Test Mode:	Transmitting	Test Engineer:	Spike Gao
Test Date:	2026-05-29	Environment:	Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa

Accumulated Transmit Time

Mode	Channel	Real Observed Period (s)	Occupancy Time for Single Hop (ms)	Hops in Observed Period	Accumulated Transmit Time (s)	Limit (s)
3DH5	Low	31.6	2.922	89	0.260	≤ 0.4
	High	31.6	2.922	87	0.254	
	*Note: Real observed period=79*400ms=31.6s*					

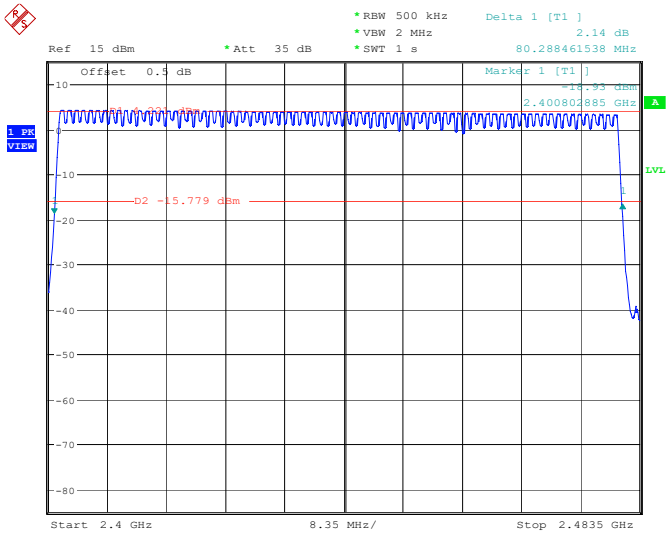
Frequency Occupation

Mode	Channel	Dwell Time (ms)	Real Observed Period (ms)	Hops in Observed Period	Limit
3DH5	Low	3.750	1185	6	≥ 1
	High	3.766	1190	3	
	*Note: Real observed period=Dwell time per hop*79*4(ms)*				

Hopping Sequence

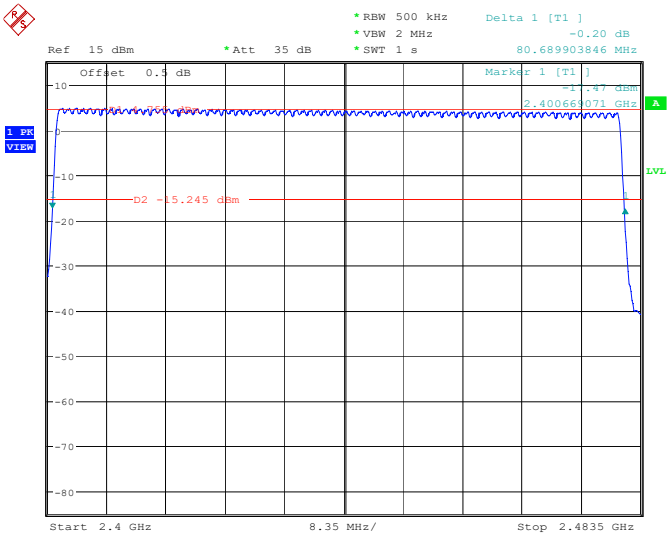
Mode	Frequency Range(MHz)	Number of Hopping Channel	Limit	20dB Occupied Bandwidth (MHz)	Limit(MHz)
GFSK	2400-2483.5	79	≥ 15	80.288	≥ 58.45
π/4-DQPSK		79		80.690	
8DPSK		79		80.690	

BDR Mode (GFSK): Number of Hopping Channels



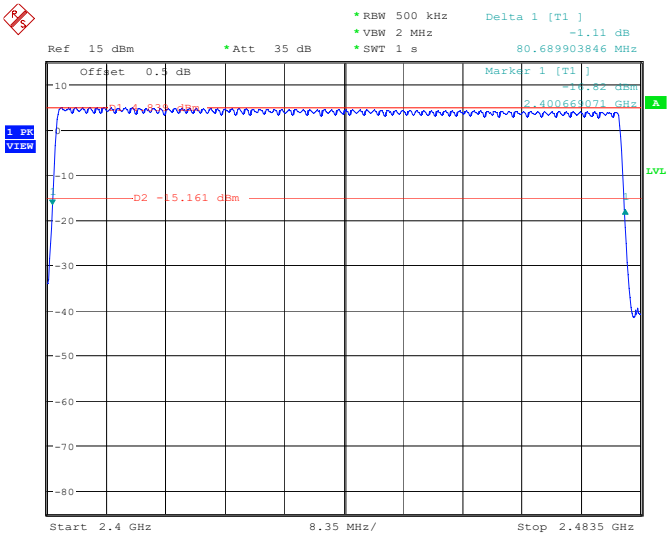
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Date: 29.MAY.2026 14:06:21

EDR Mode (π/4-DQPSK): Number of Hopping Channels



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EDR Mode (8DPSK): Number of Hopping Channels



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Date: 29.MAY.2026 14:36:41

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.5 – HOPPING FREQUENCY SEPARATION

Applicable Standard

This requirement applies to all types of FHSS equipment.

The Hopping Frequency Separation is the frequency separation between two adjacent hopping frequencies.

Limit:

For adaptive FHSS equipment, the minimum Hopping Frequency Separation shall be 100 kHz.

Adaptive Frequency Hopping equipment that switched to a non-adaptive mode for one or more hopping frequencies because interference was detected on these hopping frequencies with a level above the threshold level defined in clause 4.3.1.7.2.2, point 5 or clause 4.3.1.7.3.2, point 5, is allowed to continue to operate with a minimum Hopping Frequency Separation of 100 kHz as long as the interference remains present on these hopping frequencies. The equipment shall continue to operate in an adaptive mode on other hopping frequencies.

Adaptive FHSS equipment which decided to operate in a non-adaptive mode on one or more hopping frequencies without the presence of interference, shall comply with the limit for Hopping Frequency Separation for non-adaptive FHSS equipment defined in clause 4.3.1.5.3.1 (first paragraph) for these hopping frequencies.

Test Procedure

Option 1, the test procedure shall be as follows:

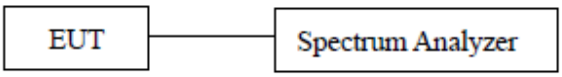
Step 1:

- The output of the transmitter shall be connected to a spectrum analyzer or equivalent.
- The analyzer shall be set as follows:
  - Centre Frequency: Centre of the two adjacent hopping frequencies
  - Frequency Span: Sufficient to see the complete power envelope of both hopping frequencies
  - RBW: 1 % of the Span
  - VBW: 3 * RBW
  - Detector Mode: Max Peak
  - Trace Mode: Max Hold
  - Sweep Time: Auto

Step 2:

- Wait for the trace to stabilize.
- Use the marker-delta function to determine the Hopping Frequency Separation between the centres of the two adjacent hopping frequencies (e.g. by identifying peaks or notches at the centre of the power envelope for the two adjacent signals). This value shall be compared with the limits defined in clause 4.3.1.5.3 and shall be recorded in the test report.

Test Setup Block diagram



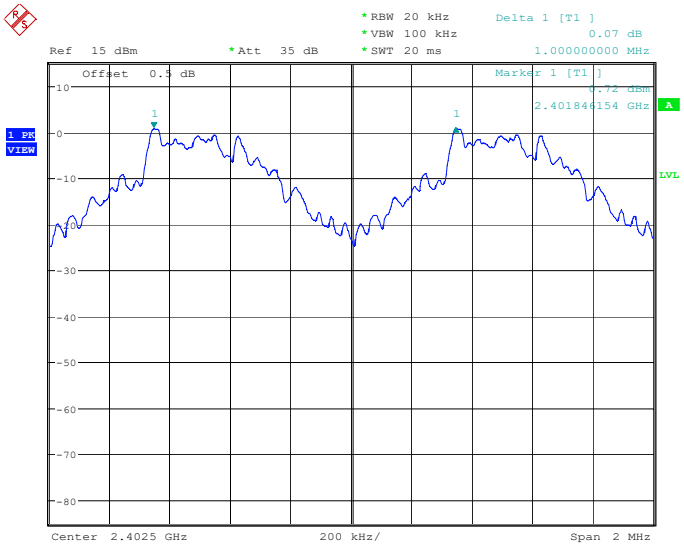
Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

Test Data

Test Mode:	Transmitting	Test Engineer:	Spike Gao
Test Date:	2026-05-29	Environment:	Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa

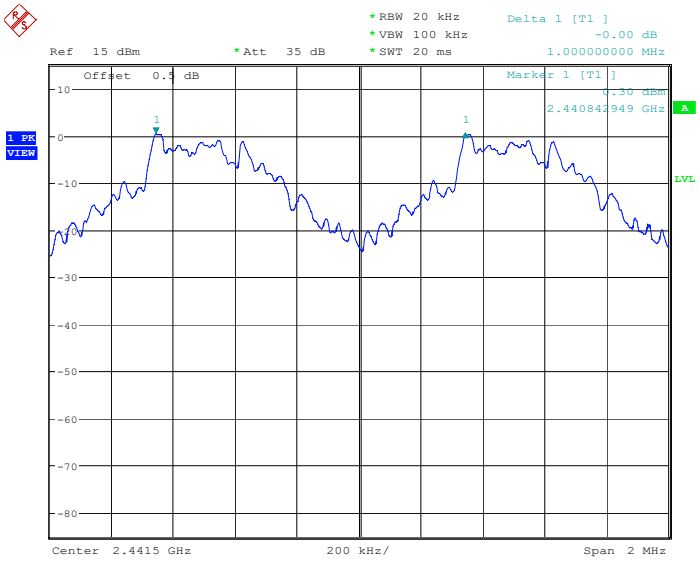
Mode	Channel	Channel Frequency (MHz)	Result (MHz)	Limit (MHz)
GFSK	Lowest	2402-2403	1.000	≥0.1
	Middle	2441-2442	1.000	
	Highest	2480-2479	1.003	
π/4-DQPSK	Lowest	2402-2403	1.000	
	Middle	2441-2442	1.000	
	Highest	2480-2479	1.003	
8DPSK	Lowest	2402-2403	1.000	
	Middle	2441-2442	1.000	
	Highest	2480-2479	1.000	

GFSK_ Lowest _DH1



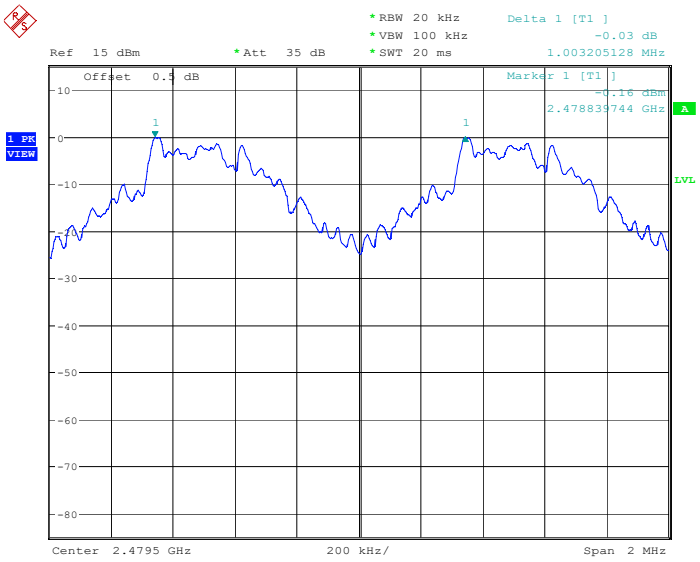
ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:24:30

GFSK_ Middle _DH1



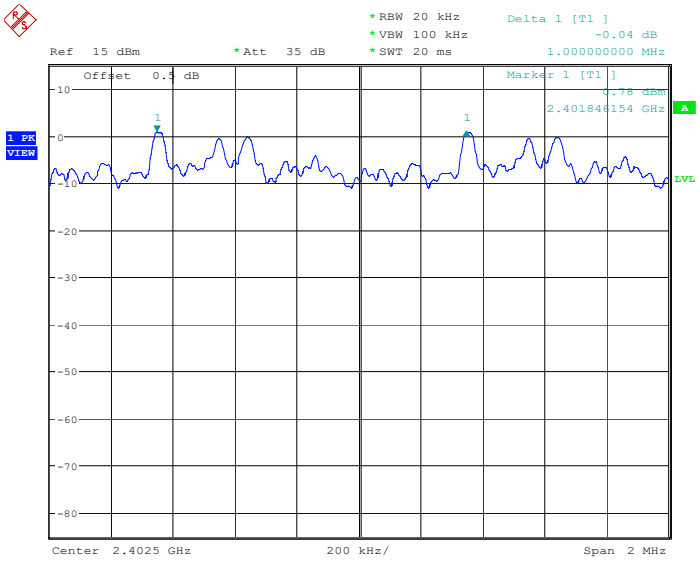
ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:27:17

GFSK_ Highest _DH1



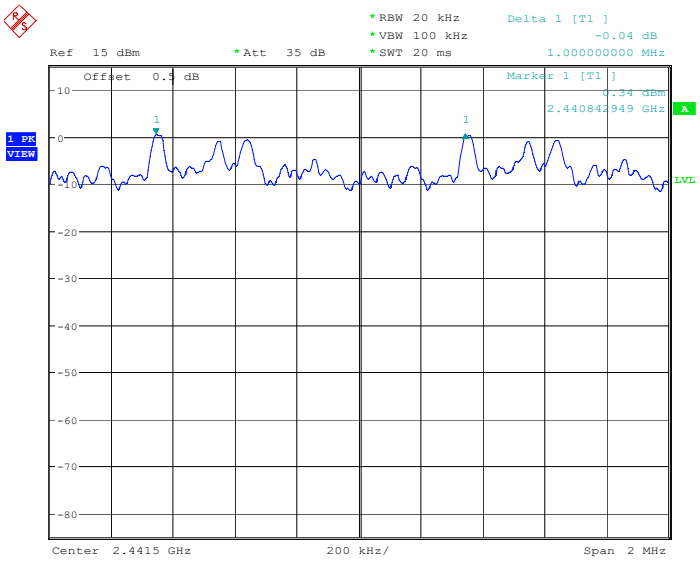
ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:30:21

π/4-DQPSK_ Lowest _2DH1



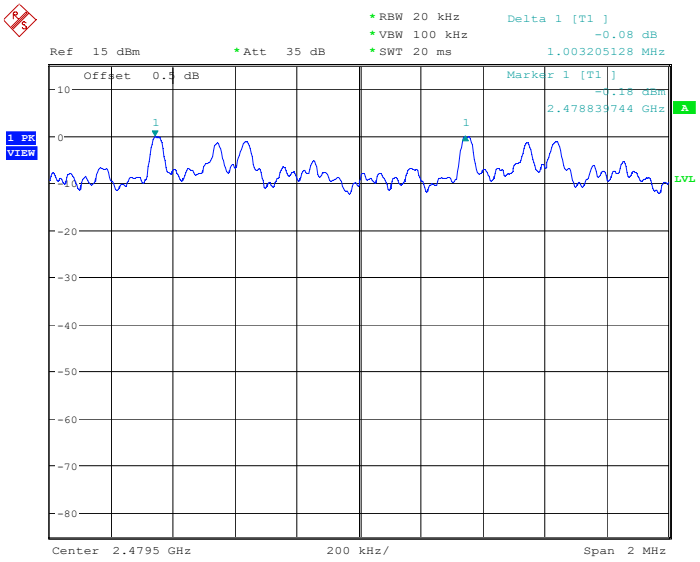
ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:33:11

π/4-DQPSK_ Middle _2DH1



ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:43:38

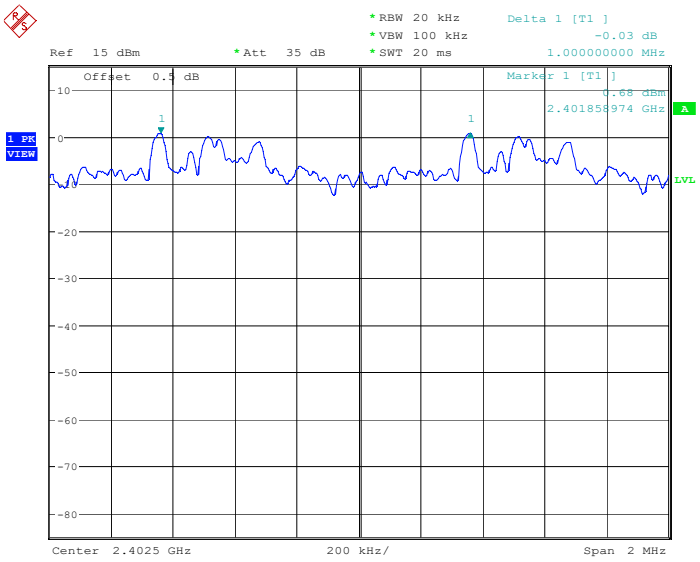
π/4-DQPSK_ Highest _2DH1



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 13:47:11

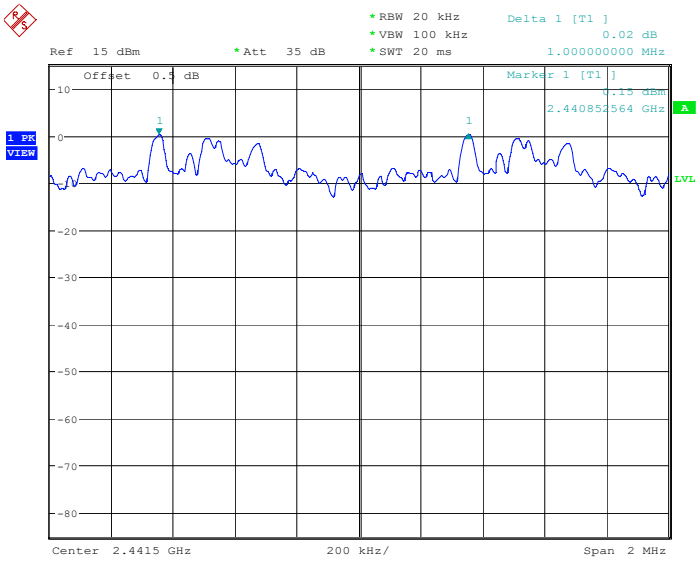
8DPSK_ Lowest _3DH1



ProjectNo.:2607T30160E-RF Tester:Spike Gao

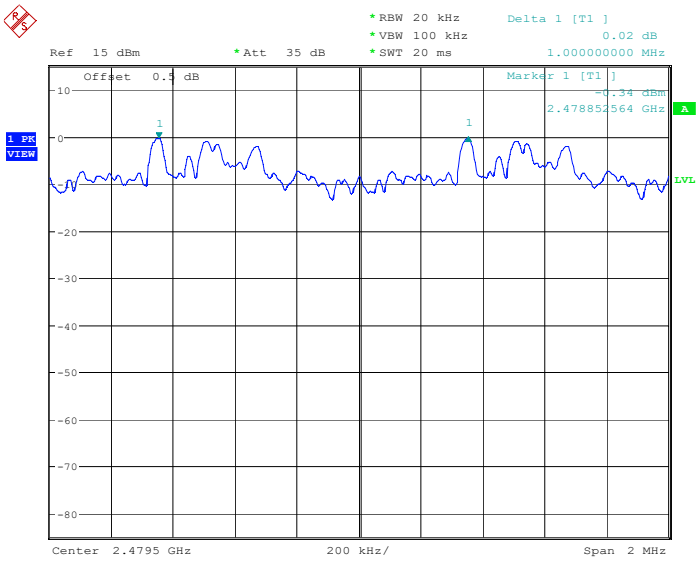
Date: 29.MAY.2026 13:50:58

8DPSK_ Middle _3DH1



ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:54:23

8DPSK_ Highest _3DH1



ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 13:56:17

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.8 – OCCUPIED CHANNEL BANDWIDTH

Applicable Standard

This requirement applies to all types of FHSS equipment

The Occupied Channel Bandwidth is the bandwidth that contains 99 % of the power of the signal when considering a single hopping frequency.

Limit:

The Occupied Channel Bandwidth for each hopping frequency shall fall completely within the band given in table 1.

Test Procedure

The measurement procedure shall be as follows:

Step 1:

Connect the UUT to the spectrum analyser and use the following settings:

- Centre Frequency: The centre frequency of the channel under test
- Resolution BW: ~ 1 % of the span without going below 1 %
- Video BW: 3 * RBW
- Frequency Span: 2 * Nominal Channel Bandwidth
- Detector Mode: RMS
- Trace Mode: Max Hold
- Sweep time: 1 s

Step 2:

Wait for the trace to stabilize.

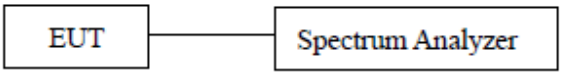
Find the peak value of the trace and place the analyser marker on this peak.

Step 3:

Use the 99 % bandwidth function of the spectrum analyser to measure the Occupied Channel Bandwidth of the UUT. This value shall be recorded.

NOTE: Make sure that the power envelope is sufficiently above the noise floor of the analyser to avoid the noise signals left and right from the power envelope being taken into account by this measurement.

Test Setup Block diagram

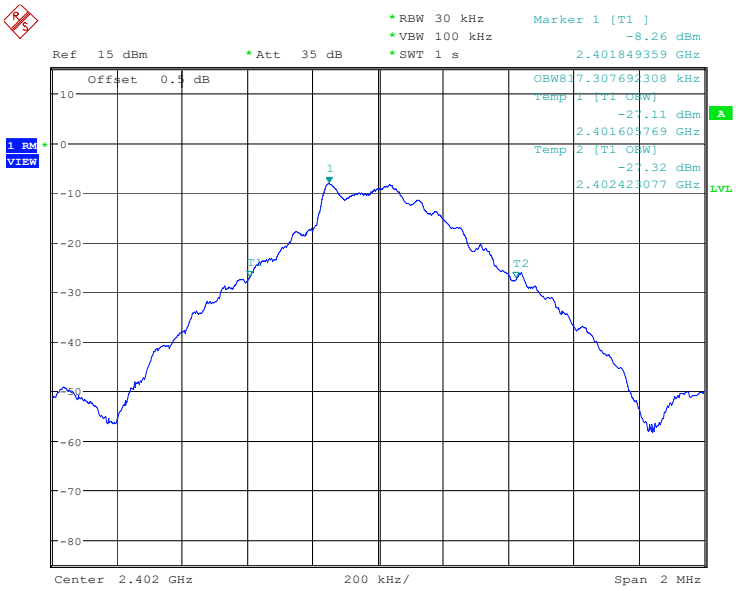


Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

Test Data

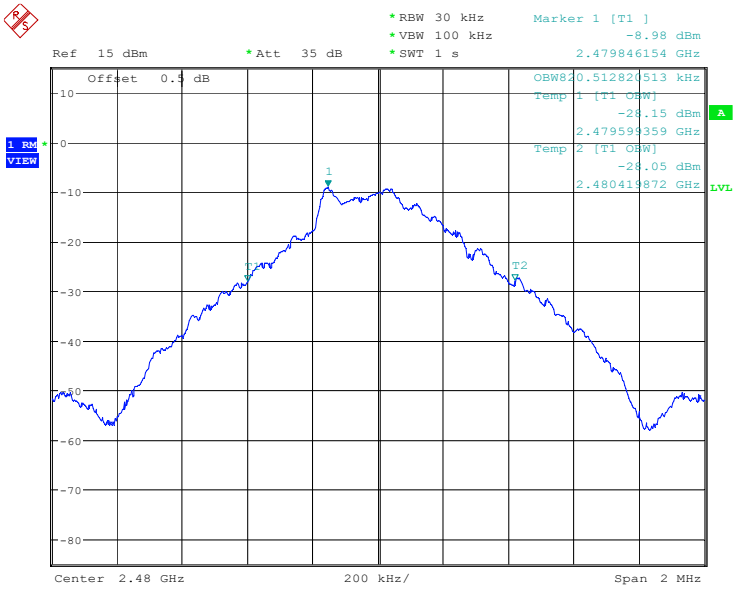
Test Mode:	Transmitting	Test Engineer:	Spike Gao
Test Date:	2026-05-29	Environment:	Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa
Mode	**Channel**	**Frequency (MHz)**	**Result (MHz)**
GFSK	Lowest	2402	0.817
	Highest	2480	0.821
π/4-DQPSK	Lowest	2402	1.163
	Highest	2480	1.160
8DPSK	Lowest	2402	1.167
	Highest	2480	1.167

BDR Mode (GFSK): Lowest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao
Date: 29.MAY.2026 14:44:20

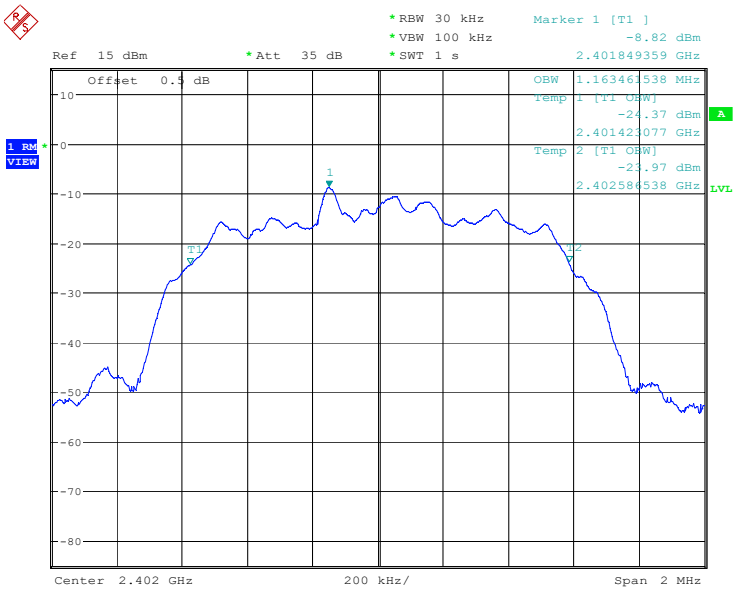
BDR Mode (GFSK): Highest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 14:45:50

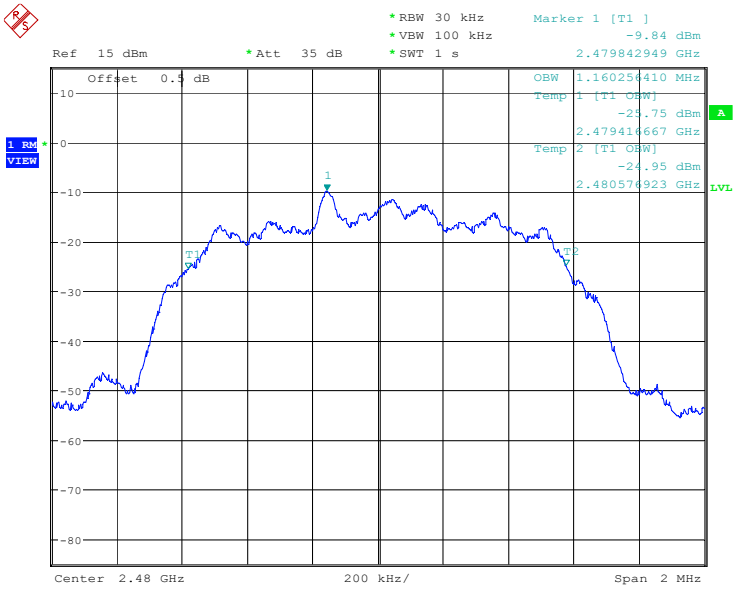
EDR Mode (π/4-DQPSK): Lowest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 14:48:30

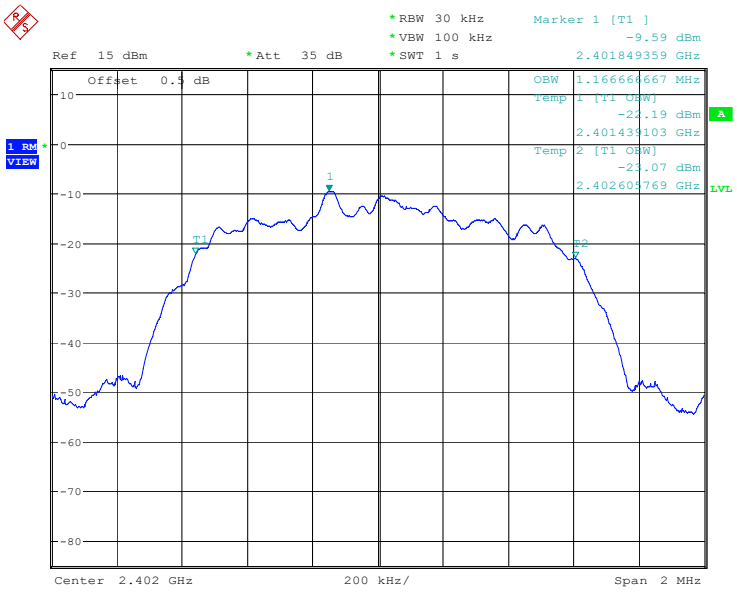
EDR Mode (π/4-DQPSK): Highest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 14:50:03

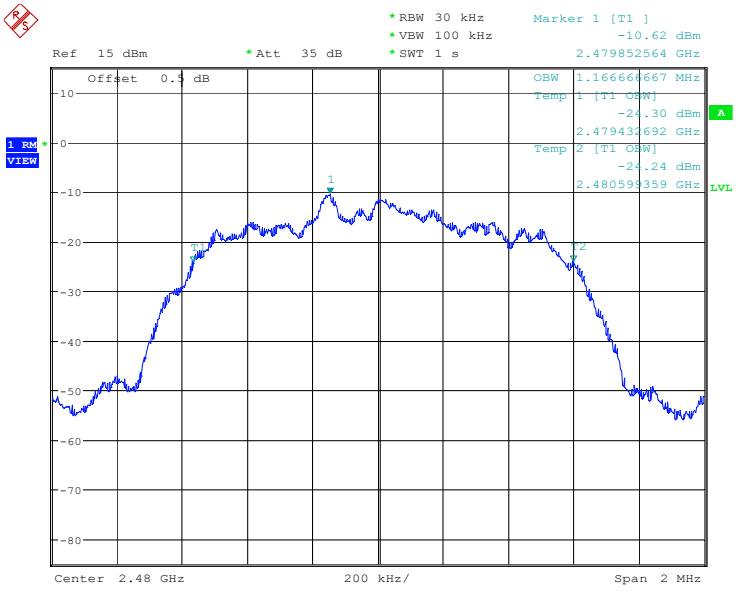
EDR Mode (8DPSK): Lowest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 14:51:14

EDR Mode (8DPSK): Highest Channel



ProjectNo.:2607T30160E-RF Tester:Spike Gao

Date: 29.MAY.2026 14:51:58

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.9 – TRANSMITTER UNWANTED EMISSION IN THE OUT-OF-BAND DOMAIN

Applicable Standard

This requirement applies to all types of FHSS equipment.

Transmitter unwanted emissions in the out-of-band domain are emissions when the equipment is in Transmit mode, on frequencies immediately outside the necessary bandwidth which results from the modulation process, but excluding spurious emissions.

Limit:

The transmitter unwanted emissions in the out-of-band domain but outside the allocated band, shall not exceed the values provided by the mask in figure 1.

Within the band specified in table 1, the Out-of-band emissions are fulfilled by compliance with the Occupied Channel Bandwidth requirement in clause 4.3.1.8.

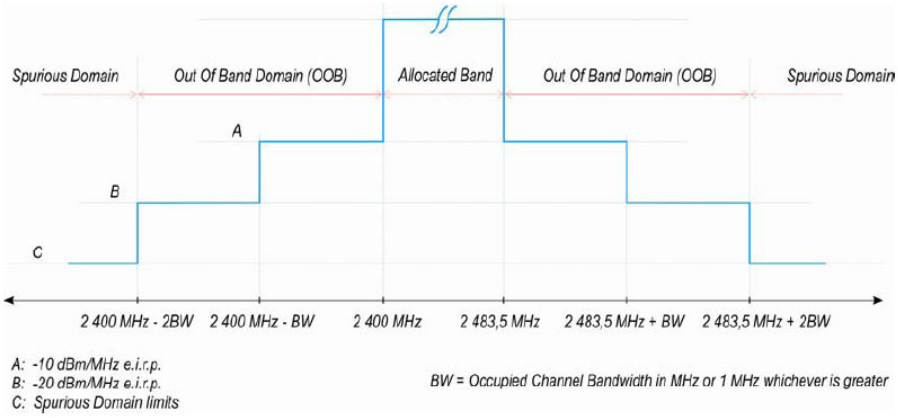


Figure 1: Transmit mask

Test Procedure

Conducted measurement:

The applicable mask is defined by the measurement results from the tests performed under clause 5.4.7 (Occupied Channel Bandwidth).

The Out-of-band emissions within the different horizontal segments of the mask provided in figures 1 and 3 shall be measured using the procedure in step 1 to step 6 below. This method assumes the spectrum analyser is equipped with the Time Domain Power option.

Step 1:

- Connect the UUT to the spectrum analyser and use the following settings:
  - Centre Frequency: 2 484 MHz
  - Span: 0 Hz

- Resolution BW: 1 MHz
- Filter mode: Channel filter
- Video BW: 3 MHz
- Detector Mode: RMS
- Trace Mode: Max Hold
- Sweep Mode: Continuous
- Sweep Points: Sweep Time [s] / (1 µs) or 5 000 whichever is greater
- Trigger Mode: Video trigger
  NOTE 1: In case video triggering is not possible, an external trigger source may be used.
- Sweep Time: > 120 % of the duration of the longest burst detected during the measurement of the RF Output Power.

Step 2: (segment 2 483,5 MHz to 2 483,5 MHz + BW)

- Adjust the trigger level to select the transmissions with the highest power level.
- For frequency hopping equipment operating in a normal hopping mode, the different hops will result in signal bursts with different power levels. In this case the burst with the highest power level shall be selected.
- Set a window (start and stop lines) to match with the start and end of the burst and in which the RMS power shall be measured using the Time Domain Power function.
- Select RMS power to be measured within the selected window and note the result which is the RMS power within this 1 MHz segment (2 483,5 MHz to 2 484,5 MHz). Compare this value with the applicable limit provided by the mask.
- Increase the centre frequency in steps of 1 MHz and repeat this measurement for every 1 MHz segment within the range 2 483,5 MHz to 2 483,5 MHz + BW. The centre frequency of the last 1 MHz segment shall be set to 2 483,5 MHz + BW - 0,5 MHz (which means this may partly overlap with the previous 1 MHz segment).

Step 3: (segment 2 483,5 MHz + BW to 2 483,5 MHz + 2BW)

- Change the centre frequency of the analyser to 2 484 MHz + BW and perform the measurement for the first 1 MHz segment within range 2 483,5 MHz + BW to 2 483,5 MHz + 2BW. Increase the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 483,5 MHz + 2 BW - 0,5 MHz. (which means this may partly overlap with the previous 1 MHz segment).

Step 4: (segment 2 400 MHz - BW to 2 400 MHz)

- Change the centre frequency of the analyser to 2 399,5 MHz and perform the measurement for the first 1 MHz segment within range 2 400 MHz - BW to 2 400 MHz Reduce the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 400 MHz - 2BW + 0,5 MHz. (which means this may partly overlap with the previous 1 MHz segment).

Step 5: (segment 2 400 MHz - 2BW to 2 400 MHz - BW)

- Change the centre frequency of the analyser to 2 399,5 MHz - BW and perform the measurement for the first 1 MHz segment within range 2 400 MHz - 2BW to 2 400 MHz - BW. Reduce the centre frequency in 1 MHz steps and repeat the measurements to cover this whole range. The centre frequency of the last 1 MHz segment shall be set to 2 400 MHz - 2BW + 0,5 MHz. (which means this may partly overlap with the previous 1 MHz segment).

Step 6:

- In case of conducted measurements on equipment with a single transmit chain, the declared antenna assembly gain "G" in dBi shall be added to the results for each of the 1 MHz segments and compared with the limits provided by the mask given in figures 1 or 3. If more than one antenna assembly is intended for this power setting, the antenna with the highest gain shall be considered.
- In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), the measurements need to be repeated for each of the active transmit chains. The declared antenna assembly gain "G" in dBi for a single antenna shall be added to these results. If more than one antenna assembly is intended for this power setting, the antenna with the highest gain shall be considered. Comparison with the applicable limits shall be done using any of the options given below:

  - Option 1: the results for each of the transmit chains for the corresponding 1 MHz segments shall be added. The additional beamforming gain "Y" in dB shall be added as well and the resulting values compared with the limits provided by the mask given in figures 1 or 3.

  - Option 2: the limits provided by the mask given in figures 1 or 3 shall be reduced by 10 x $\log_{10}(A_{ch})$ and the additional beamforming gain "Y" in dB. The results for each of the transmit chains shall be individually compared with these reduced limits.

NOTE 2: A_{ch} refers to the number of active transmit chains.

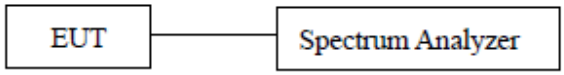
It shall be recorded whether the equipment complies with the mask provided in figures 1 or 3.

Radiated measurement:

The test set up as described in annex B and the applicable measurement procedures described in annex C shall be used.Alternatively a test fixture may be used.

The test procedure is as described under clause 5.4.8.2.1.

Test Setup Block diagram

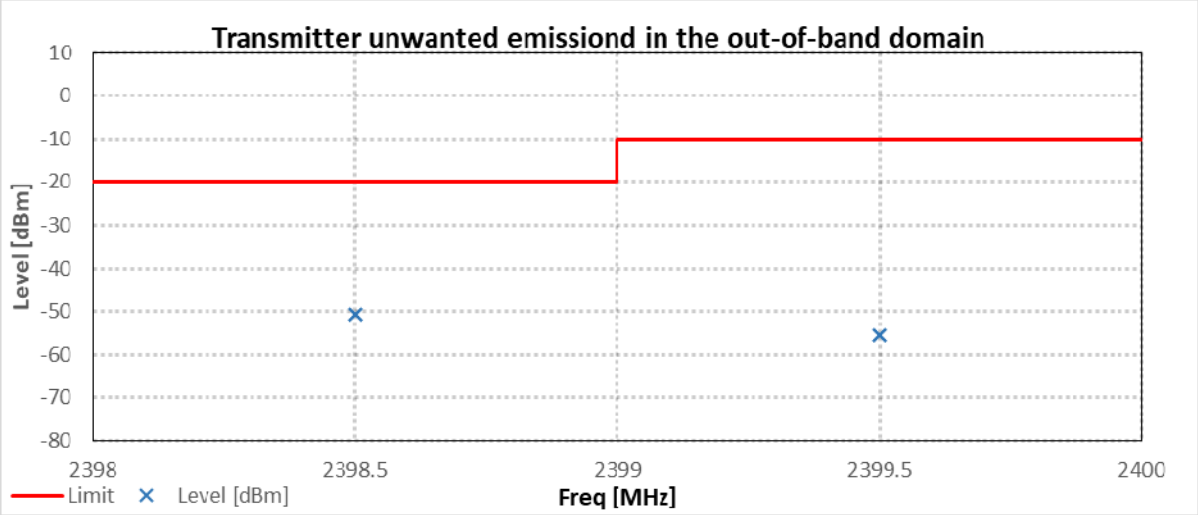


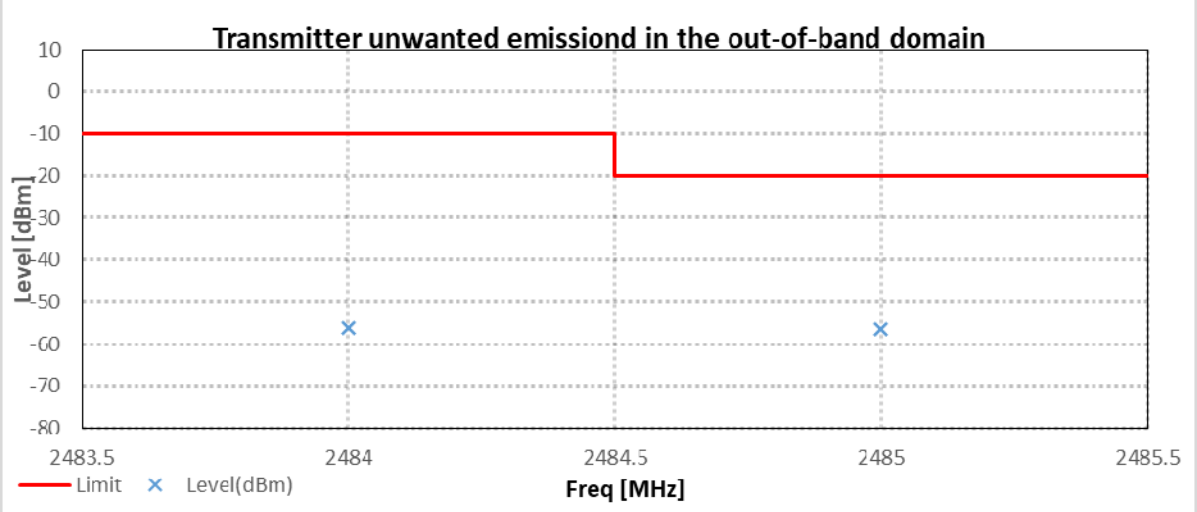
Note: The cable assembly insertion loss of 0.5dB was entered as an offset in the spectrum analyzer/power sensor. (Actual cable loss was unavailable at the time of testing, therefore loss of 0.5dB was assumed as worst case.) This was later verified to be true by laboratory.

Test Data

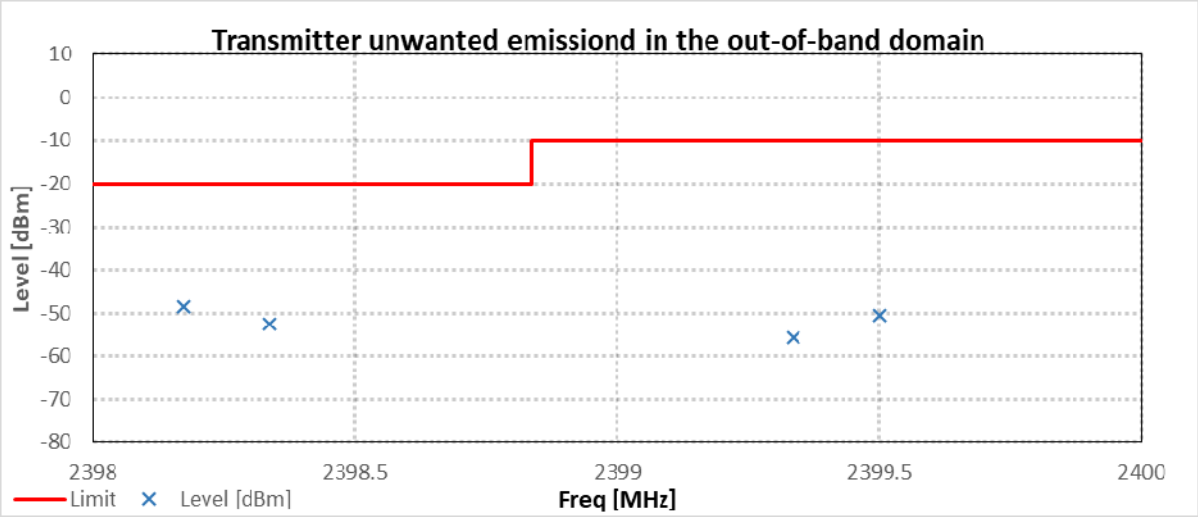
Test Mode:	Transmitting	Test Engineer:	Spike Gao
Test Date:	2026-05-29	Environment:	Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa

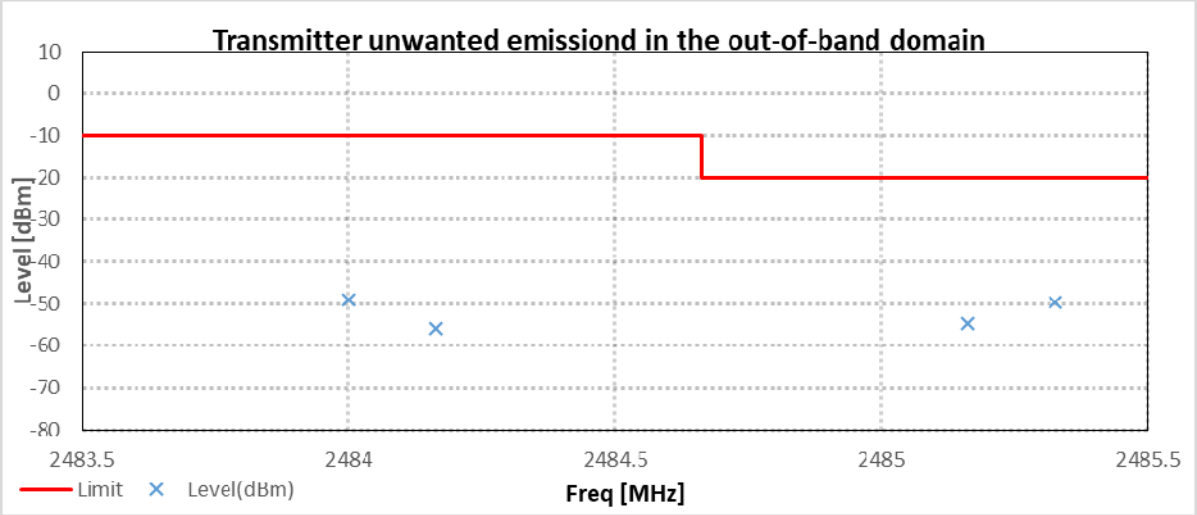
GFSK_Hopping_DH1



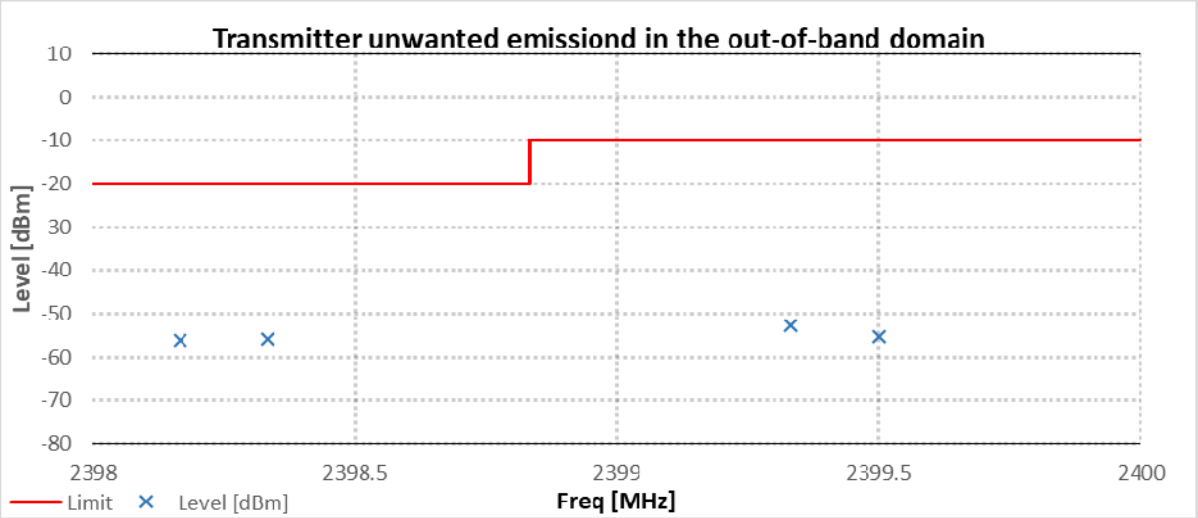


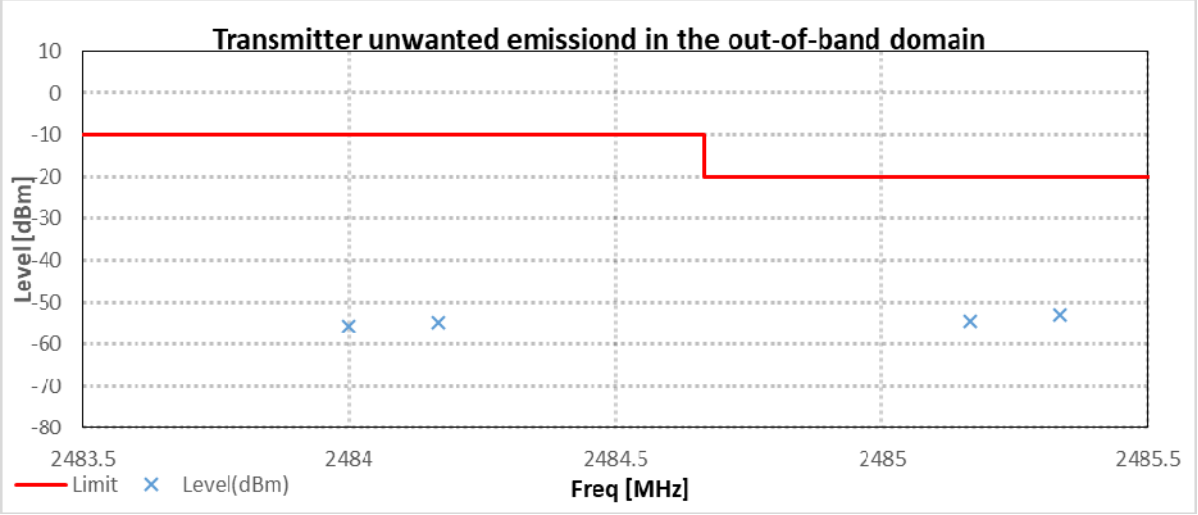
π/4-DQPSK_Hopping_2DH1





8DPSK_Hopping_3DH1





ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.10 – TRANSMITTER UNWANTED EMISSION IN THE SPURIOUS DOMAIN

Applicable Standard

Transmitter unwanted emissions in the spurious domain are emissions outside the allocated band and outside the out-of-band domain as indicated in figure 1 when the equipment is in Transmit mode.

In case of equipment with antenna connectors, these limits apply to emissions at the antenna port (conducted). For emissions radiated by the cabinet or emissions radiated by integral antenna equipment (without antenna connectors), these limits are e.r.p. for emissions up to 1 GHz and e.i.r.p. for emissions above 1 GHz.

The spurious emissions of the transmitter shall not exceed the values in the following table:

Transmitter limits for spurious emissions

Frequency Range	Maximum power e.r.p (≤ 1 GHz) e.i.r.p (> 1 GHz)	Bandwidth
30 MHz to 47 MHz	-36 dBm	100 kHz
47 MHz to 74 MHz	-54 dBm	100 kHz
74 MHz to 87.5 MHz	-36 dBm	100 kHz
87.5 MHz to 118 MHz	-54 dBm	100 kHz
118 MHz to 174 MHz	-36 dBm	100 kHz
174 MHz to 230 MHz	-54 dBm	100 kHz
230 MHz to 470 MHz	-36 dBm	100 kHz
470 MHz to 694 MHz	-54 dBm	100 kHz
694 MHz to 1 GHz	-36 dBm	100 kHz
1 GHz to 12.75 GHz	-30 dBm	1 MHz

Test Procedure

Conducted measurement:

The spectrum in the spurious domain (see figure 1 or figure 3) shall be searched for emissions that exceed the limit values given in table 4 or table 12 or that come to within 6 dB below these limits. Each occurrence shall be recorded.

The measurement procedure contains 2 parts.

Pre-scan:

The procedure in step 1 to step 4 below shall be used to identify potential unwanted emissions of the UUT.

Step 1:

The sensitivity of the measurement set-up should be such that the noise floor is at least 12 dB below the limits given in table 4 or table 12.

Step 2:

The emissions over the range 30 MHz to 1 000 MHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 100 kHz
- Video bandwidth: 300 kHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 19 400; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented
- Sweep time: -For non continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 100 kHz frequency step, the measurement time is greater than two transmissions of the UUT, on any channel
  -For Frequency Hopping equipment operating in a normal operating (hopping not disabled) mode, the sweep time shall be further increased to capture multiple transmissions on any of the hopping frequencies
  -The above sweep time setting may result in long measuring times in case of frequency hopping equipment. To avoid such long measuring times, an FFT analyser may be used

Allow the trace to stabilize. Any emissions identified during the sweeps above and that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.9.2.1.3 and compared to the limits given in table 4 or table 12.

Step 3:

The emissions over the range 1 GHz to 12,75 GHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 1 MHz
- Video bandwidth: 3 MHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 23 500; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented
- Sweep time: - For non continuous transmissions (duty cycle less than 100 %), the sweep time shall be sufficiently long, such that for each 1 MHz frequency step, the measurement time is greater than two transmissions of the UUT, on any channel
  - For Frequency Hopping equipment operating in a normal operating (hopping not disabled) mode, the sweep time shall be further increased to capture multiple transmissions on any of the hopping frequencies
  - The above sweep time setting may result in long measuring times in case of frequency hopping equipment. To avoid such long measuring times, an FFT analyser may be used

Allow the trace to stabilize. Any emissions identified during the sweeps above that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.9.2.1.3 and compared to the limits given in table 4 or table 12.

Frequency Hopping equipment may generate a block (or several blocks) of spurious emissions anywhere within the spurious domain. If this is the case, only the highest peak of each block of emissions shall be measured using the procedure in clause 5.4.9.2.1.3.

Step 4:

In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), step 2 and step 3 need to be repeated for each of the active transmit chains (Ach).The limits used to identify emissions during this pre-scan need to be reduced by $10 \times \log_{10} (A_{ch})$.

Measurement of the emissions identified during the pre-scan:

The procedure in step 1 to step 4 below shall be used to accurately measure the individual unwanted emissions identified during the pre-scan measurements above. This method assumes the spectrum analyser has a Time Domain Power function.

Step 1:

The level of the emissions shall be measured using the following spectrum analyser settings:

- Measurement Mode: Time Domain Power
- Centre Frequency: Frequency of the emission identified during the pre-scan
- Resolution Bandwidth: 100 kHz (< 1 GHz) / 1 MHz (> 1 GHz)
- Video Bandwidth: 300 kHz (< 1 GHz) / 3 MHz (> 1 GHz)
- Frequency Span: Zero Span
- Sweep mode: Single Sweep
- Sweep time: > 120 % of the duration of the longest burst detected during the measurement of the RF Output Power
- Sweep points: Sweep time [µs] / (1 µs) with a maximum of 30 000
- Trigger: Video (burst signals) or Manual (continuous signals)
- Detector: RMS

Allow the trace to stabilize. Any emissions identified during the sweeps above and that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.9.2.1.3 and compared to the limits given in table 4 or table 12.

Step 2:

Set a window where the start and stop indicators match the start and end of the burst with the highest level and record the value of the power measured within this window. If the spurious emission to be measured is a continuous transmission, the measurement window shall be set to match the start and stop times of the sweep.

Step 3:

In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), step 2 needs to be repeated for each of the active transmit chains (A_{ch}).

Sum the measured power (within the observed window) for each of the active transmit chains.

Step 4:

The value defined in step 3 shall be compared to the limits defined in table 4 or table 12.

Radiated measurement:

The test site as described in annex B and applicable measurement procedures as described in annex C shall be used.

The test procedure is further as described under clause 5.4.9.2.1.

Test Data

Test result: Compliant, please refer to following tables.

Test Mode:	Transmitting	Test Engineer:	Wlif Wu
Test Date:	2026-06-24~2026-06-26	Environment:	Temp.: 22.7~23.2℃ Humi.: 59~60% Atm.: 100.1kPa

30 MHz - 12.75 GHz								
Frequency (MHz)	Polar (H/V)	Receiver Reading (dBμV)	Substituted Method			Absolute Level (dBm)	Limit (dBm)	Margin (dB)
			Substituted Level (dBm)	Antenna Gain (dBd/dBi)	Cable Loss (dB)			
BDR DH5_low channel (2402 MHz)								
225.21	H	30.60	-70.17	0.00	0.27	-70.44	-54.00	16.44
64.29	V	29.55	-83.19	-8.03	0.15	-91.37	-54.00	37.37
806.63	H	30.38	-69.07	0.00	0.41	-69.48	-36.00	33.48
173.42	V	32.84	-76.63	0.00	0.24	-76.87	-36.00	40.87
4804.000	H	61.55	-44.04	11.80	2.69	-34.93	-30.00	4.93
4804.000	V	53.65	-52.24	11.80	2.69	-43.13	-30.00	13.13
7206.000	H	49.64	-51.30	9.00	3.29	-45.59	-30.00	15.59
7206.000	V	46.69	-54.97	9.00	3.29	-49.26	-30.00	19.26
BDR DH5_high channel (2480 MHz)								
400.88	H	29.83	-76.41	0.00	0.33	-76.74	-36.00	40.74
173.22	V	31.43	-78.06	0.00	0.24	-78.30	-36.00	42.30
711.28	H	30.71	-65.47	0.00	0.40	-65.87	-36.00	29.87
713.61	V	31.28	-65.06	0.00	0.40	-65.46	-36.00	29.46
4960.000	H	62.38	-43.28	10.68	2.72	-35.32	-30.00	5.32
4960.000	V	55.70	-50.43	10.68	2.72	-42.47	-30.00	12.47
7440.000	H	49.74	-50.89	7.86	3.33	-46.36	-30.00	16.36
7440.000	V	47.19	-54.49	7.86	3.33	-49.96	-30.00	19.96

30 MHz - 12.75 GHz								
Frequency (MHz)	Polar (H/V)	Receiver Reading (dBμV)	Substituted Method			Absolute Level (dBm)	Limit (dBm)	Margin (dB)
			Substituted Level (dBm)	Antenna Gain (dBd/dBi)	Cable Loss (dB)			
EDR 2DH5_low channel (2402 MHz)								
148.58	H	29.63	-82.19	0.00	0.23	-82.42	-36.00	46.42
173.22	V	31.81	-77.68	0.00	0.24	-77.92	-36.00	41.92
267.31	H	30.61	-68.62	0.00	0.28	-68.90	-36.00	32.90
742.81	V	31.25	-67.80	0.00	0.41	-68.21	-36.00	32.21
4804.000	H	62.71	-42.88	11.80	2.69	-33.77	-30.00	3.77
4804.000	V	52.81	-53.08	11.80	2.69	-43.97	-30.00	13.97
7206.000	H	48.65	-52.29	9.00	3.29	-46.58	-30.00	16.58
7206.000	V	45.94	-55.72	9.00	3.29	-50.01	-30.00	20.01
EDR 2DH5_high channel (2480 MHz)								
326.58	H	29.92	-78.63	0.00	0.30	-78.93	-36.00	42.93
173.12	V	33.90	-75.60	0.00	0.24	-75.84	-36.00	39.84
632.71	H	30.54	-68.77	0.00	0.39	-69.16	-54.00	15.16
244.23	V	30.52	-64.71	0.00	0.28	-64.99	-36.00	28.99
4960.000	H	64.38	-41.28	10.68	2.72	-33.32	-30.00	3.32
4960.000	V	57.99	-48.14	10.68	2.72	-40.18	-30.00	10.18
7440.000	H	49.05	-51.58	7.86	3.33	-47.05	-30.00	17.05
7440.000	V	48.01	-53.67	7.86	3.33	-49.14	-30.00	19.14
EDR 3DH5_low channel (2402 MHz)								
90.67	H	31.17	-84.11	0.00	0.18	-84.29	-54.00	30.29
55.95	V	30.57	-80.49	-12.16	0.15	-92.80	-54.00	38.80
159.84	H	28.39	-82.52	0.00	0.23	-82.75	-36.00	46.75
172.45	V	34.22	-75.33	0.00	0.24	-75.57	-36.00	39.57
4804.000	H	63.64	-41.95	11.80	2.69	-32.84	-30.00	2.84
4804.000	V	53.90	-51.99	11.80	2.69	-42.88	-30.00	12.88
7206.000	H	49.72	-51.22	9.00	3.29	-45.51	-30.00	15.51
7206.000	V	47.35	-54.31	9.00	3.29	-48.60	-30.00	18.60
EDR 3DH5_high channel (2480 MHz)								
149.07	H	30.14	-81.66	0.00	0.23	-81.89	-36.00	45.89
173.42	V	31.66	-77.81	0.00	0.24	-78.05	-36.00	42.05
408.06	H	30.48	-74.49	0.00	0.33	-74.82	-36.00	38.82
219.88	V	30.79	-71.16	0.00	0.26	-71.42	-54.00	17.42
4960.000	H	64.66	-41.00	10.68	2.72	-33.04	-30.00	3.04
4960.000	V	57.35	-48.78	10.68	2.72	-40.82	-30.00	10.82
7440.000	H	50.76	-49.87	7.86	3.33	-45.34	-30.00	15.34
7440.000	V	48.00	-53.68	7.86	3.33	-49.15	-30.00	19.15

Note 1: The unit of antenna gain is dBd for frequency below 1GHz and is dBi for frequency above 1GHz.
Note 2: Absolute Level = Substituted Level - Cable loss + Antenna Gain
Margin = Limit- Absolute Level

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.11 - RECEIVER SPURIOUS EMISSIONS

Applicable Standard

Receiver spurious emissions are emissions at any frequency when the equipment is in receive mode.

Limit:

The spurious emissions of the receiver shall not exceed the values given in table 5.

In case of equipment with antenna connectors, these limits apply to emissions at the antenna port (conducted). For emissions radiated by the cabinet or emissions radiated by integral antenna equipment (without antenna connectors), these limits are e.r.p. for emissions up to 1 GHz and e.i.r.p. for emissions above 1 GHz.

Table 5: Spurious emission limits for receivers

Frequency range	Maximum power e.r.p.((≤ 1 GHz) e.i.r.p. (> 1 GHz	Measurement bandwith
30 MHz to 1GHz	-57 dBm	100 kHz
1 GHz to 12.75GHz	-47 dBm	1 MHz

Test Procedure

Conducted measurement:

In case of conducted measurements, the radio equipment shall be connected to the measuring equipment via an attenuator.
The spectrum in the spurious domain (see figure 1 or figure 3) shall be searched for emissions that exceed the limit values given in table 5 or table 13 or that come to within 6 dB below these limits. Each occurrence shall be recorded.

The measurement procedure contains 2 parts.

Pre-scan:

The procedure in step 1 to step 4 below shall be used to identify potential unwanted emissions of the UUT.

Step 1:

The sensitivity of the measurement set-up should be such that the noise floor is at least 12 dB below the limits given in table 5 or table 13.

Step 2:

The emissions over the range 30 MHz to 1 000 MHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 100 kHz
- Video bandwidth: 300 kHz

- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 19 400
- Sweep time: Auto

Wait for the trace to stabilize. Any emissions identified during the sweeps above and that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.10.2.1.3 and compared to the limits given in table 5 or table 13.

Step 3:

The emissions over the range 1 GHz to 12,75 GHz shall be identified.

Spectrum analyser settings:

- Resolution bandwidth: 1 MHz
- Video bandwidth: 3 MHz
- Filter type: 3 dB (Gaussian)
- Detector mode: Peak
- Trace Mode: Max Hold
- Sweep Points: ≥ 23 500; for spectrum analysers not supporting this high number of sweep points, the frequency band may be segmented
- Sweep time: Auto

Wait for the trace to stabilize. Any emissions identified during the sweeps above that fall within the 6 dB range below the applicable limit or above, shall be individually measured using the procedure in clause 5.4.10.2.1.3 and compared to the limits given in table 5 or table 13.

Frequency Hopping equipment may generate a block (or several blocks) of spurious emissions anywhere within the spurious domain. If this is the case, only the highest peak of each block of emissions shall be measured using the procedure in clause 5.4.10.2.1.3.

Step 4:

In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), step 2 and step 3 need to be repeated for each of the active transmit chains (A_{ch}).The limits used to identify emissions during this pre-scan need to be reduced by $10 \times \log_{10} (A_{ch})$.

Measurement of the emissions identified during the pre-scan:

The procedure in step 1 to step 4 below shall be used to accurately measure the individual unwanted emissions identified during the pre-scan measurements above. This method assumes the spectrum analyser has a Time Domain Power function.

Step 1:

The level of the emissions shall be measured using the following spectrum analyser settings:

- Measurement Mode: Time Domain Power
- Centre Frequency: Frequency of the emission identified during the pre-scan
- Resolution Bandwidth: 100 kHz (< 1 GHz) / 1 MHz (> 1 GHz)
- Video Bandwidth: 300 kHz (< 1 GHz) / 3 MHz (> 1 GHz)
- Frequency Span: Zero Span

- Sweep mode: Single Sweep
- Sweep time: 30 ms
- Sweep points: ≥ 30 000
- Trigger: Video (for burst signals) or Manual (for continuous signals)
- Detector: RMS

Step 2:

Set a window where the start and stop indicators match the start and end of the burst with the highest level and record the value of the power measured within this window. If the spurious emission to be measured is a continuous transmission, the measurement window shall be set to match the start and stop times of the sweep.

Step 3:

In case of conducted measurements on smart antenna systems (equipment with multiple transmit chains), step 2 needs to be repeated for each of the active transmit chains (A_{ch}).

Sum the measured power (within the observed window) for each of the active transmit chains.

Step 4:

The value defined in step 3 shall be compared to the limits defined in table 5 or table 13.

Radiated measurement:

The test site as described in annex B and applicable measurement procedures as described in annex C shall be used.

The test procedure is further as described under clause 5.4.10.2.1.

Test Data

Test result: Compliant, please refer to following tables.

Test Mode:	Receiving	Test Engineer:	Wlif Wu
Test Date:	2026-06-24~2026-06-26	Environment:	Temp.: 22.7~23.2℃ Humi.: 59~60% Atm.: 100.1kPa

30 MHz - 12.75 GHz

Frequency (MHz)	Polar (H/V)	Receiver Reading (dBμV)	Substituted Method			Absolute Level (dBm)	Limit (dBm)	Margin (dB)
			Substituted Level (dBm)	Antenna Gain (dBd/dBi)	Cable Loss (dB)			
BDR DH5_low channel (2402 MHz)								
672.77	H	31.14	-65.30	0.00	0.39	-65.69	-57.00	8.69
765.50	V	30.26	-69.43	0.00	0.41	-69.84	-57.00	12.84
1817.800	H	52.16	-59.68	6.70	1.71	-54.69	-47.00	7.69
2656.750	V	53.20	-53.10	2.20	2.05	-52.95	-47.00	5.95
BDR DH5_high channel (2480 MHz)								
442.78	H	36.59	-62.21	0.00	0.35	-62.56	-57.00	5.56
71.86	V	38.51	-74.85	-4.07	0.16	-79.08	-57.00	22.08
1811.925	H	53.77	-58.15	6.70	1.70	-53.15	-47.00	6.15
1818.975	V	52.48	-59.64	6.70	1.71	-54.65	-47.00	7.65
EDR 2DH5_low channel (2402 MHz)								
808.18	H	31.53	-67.76	0.00	0.41	-68.17	-57.00	11.17
551.72	V	37.44	-67.35	0.00	0.38	-67.73	-57.00	10.73
1815.450	H	52.10	-59.77	6.70	1.71	-54.78	-47.00	7.78
1817.800	V	52.23	-59.91	6.70	1.71	-54.92	-47.00	7.92
EDR 2DH5_high channel (2480 MHz)								
797.51	H	31.17	-68.93	0.00	0.41	-69.34	-57.00	12.34
173.42	V	33.15	-76.32	0.00	0.24	-76.56	-57.00	19.56
1816.625	H	51.64	-60.22	6.70	1.71	-55.23	-47.00	8.23
1817.800	V	51.52	-60.62	6.70	1.71	-55.63	-47.00	8.63
EDR 3DH5_low channel (2402 MHz)								
575.38	H	30.64	-73.22	0.00	0.38	-73.60	-57.00	16.60
172.35	V	33.47	-76.09	0.00	0.24	-76.33	-57.00	19.33
1811.925	H	51.12	-60.80	6.70	1.70	-55.80	-47.00	8.80
1816.625	V	53.36	-58.79	6.70	1.71	-53.80	-47.00	6.80
EDR 3DH5_high channel (2480 MHz)								
826.42	H	30.63	-66.82	0.00	0.42	-67.24	-57.00	10.24
173.42	V	33.30	-76.17	0.00	0.24	-76.41	-57.00	19.41
1811.925	H	51.76	-60.16	6.70	1.70	-55.16	-47.00	8.16
1811.925	V	51.79	-60.42	6.70	1.70	-55.42	-47.00	8.42

Note 1: The unit of antenna gain is dBd for frequency below 1GHz and is dBi for frequency above 1GHz.
Note 2: Absolute Level = Substituted Level - Cable loss + Antenna Gain
Margin = Limit- Absolute Level

ETSI EN 300 328 V2.2.2 (2019-07) Clause 4.3.1.12 – RECEIVER BLOCKING

Applicable Standard

Receiver blocking is a measure of the ability of the equipment to receive a wanted signal on its operating channel without exceeding a given degradation in the presence of an unwanted signal (blocking signal) on frequencies other than those of the operating band provided in table 1.

While maintaining the minimum performance criteria as defined in clause 4.3.1.12.3, the blocking levels at specified frequency offsets shall be equal to or greater than the limits defined for the applicable receiver category provided in table 6, table 7 or table 8.

Table 6: Receiver Blocking parameters for Receiver Category 1 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 4)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 4)	Type of blocking signal
(-133 dBm + 10 × $\log_{10}$(OCBW)) or -68 dBm whichever is less (see note 2)	2 380 2 504	-34	CW
(-139 dBm + 10 × $\log_{10}$(OCBW)) or -74 dBm whichever is less (see note 3)	2 300 2 330 2 360 2 524 2 584 2 674		

NOTE 1: OCBW is in Hz.
NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 26 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal.
NOTE 3: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 20 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal.
NOTE 4: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.

Table 7: Receiver Blocking parameters receiver Category 2 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 3)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 3)	Type of blocking signal
(-139 dBm + 10 × $\log_{10}$(OCBW) + 10 dB) or (-74 dBm + 10 dB) whichever is less (see note 2)	2 380 2 504 2 300 2 584	-34	CW
NOTE 1: OCBW is in Hz. NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative test may be performed using a wanted signal up to P_{min} + 26 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal. NOTE 3: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.			

Table 8: Receiver Blocking parameters receiver Category 3 equipment

Wanted signal mean power from companion device (dBm) (see notes 1 and 3)	Blocking signal frequency (MHz)	Blocking signal power (dBm) (see note 3)	Type of blocking signal
(-139 dBm + 10 × $\log_{10}$(OCBW) + 20 dB) or (-74 dBm + 20 dB) whichever is less (see note 2)	2 380 2 504 2 300 2 584	-34	CW
NOTE 1: OCBW is in Hz. NOTE 2: In case of radiated measurements using a companion device and the level of the wanted signal from the companion device cannot be determined, a relative the test may be performed using a wanted signal up to P_{min} + 30 dB where P_{min} is the minimum level of wanted signal required to meet the minimum performance criteria as defined in clause 4.3.1.12.3 in the absence of any blocking signal. NOTE 3: The level specified is the level at the UUT receiver input assuming a 0 dBi antenna assembly gain. In case of conducted measurements, this level has to be corrected for the (in-band) antenna assembly gain (G). In case of radiated measurements, this level is equivalent to a power flux density (PFD) in front of the UUT antenna with the UUT being configured/positioned as recorded in clause 5.4.3.2.2.			

Test Procedure

According to ETSI EN 300 328 V2.2.2 (2019-07) Clause5.4.11.2

Test Setup Block diagram

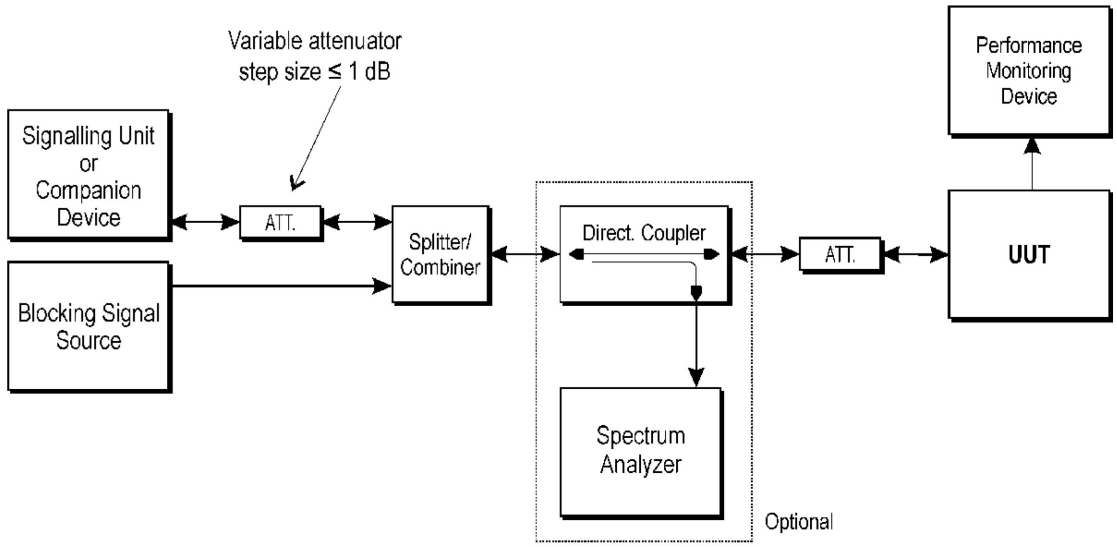


Figure 6: Test Set-up for receiver blocking

Test data

Test Mode:	Receiving		**Test Engineer:**	Spike Gao	
Test Date:	2026-05-29		**Environment:**	Temp.: 23.1℃ Humi.: 56% Atm.: 100.1kPa	
Mode	**Receiver Category**	**Channel**	**Blocking signal frequency (MHz)**	**PER (%)**	**Limit(%)**
GFSK	2	Low	2300	2.22	≤ 10
			2380	2.17	
		High	2504	1.92	
			2584	2.28	

Note: PER was monitored by CMW500.

EXHIBIT A - E.2 INFORMATION AS REQUIRED BY ETSI EN 300 328 V2.2.2, Clause 5.4.1

In accordance with EN 300 328, clause 5.4.1, the following information is provided by the manufacturer.

a) The type of modulation used by the equipment:

☒ FHSS
☐ other forms of modulation

b) In case of FHSS modulation:

In case of non-Adaptive Frequency Hopping equipment:
The number of Hopping Frequencies:_____.

In case of Adaptive Frequency Hopping Equipment:
The maximum number of Hopping Frequencies:__79__;
The minimum number of Hopping Frequencies: __79____;

The (average) Dwell Time: __3.766 ms__;

c) Adaptive / non-adaptive equipment:

☐ non-adaptive Equipment
☒ adaptive Equipment without the possibility to switch to a non-adaptive mode
☐ adaptive Equipment which can also operate in a non-adaptive mode

d) In case of adaptive equipment:

The maximum Channel Occupancy Time implemented by the equipment: _____ms

☐ The equipment has implemented an LBT based DAA mechanism

In case of equipment using modulation different from FHSS:

☐ The equipment is Frame Based equipment
☐ The equipment is Load Based equipment
☐ The equipment can switch dynamically between Frame Based and Load Based equipment

The CCA time implemented by the equipment: _____µs

☐ The equipment has implemented a non-LBT based DAA mechanism
☐ The equipment can operate in more than one adaptive mode

e) In case of non-adaptive Equipment:

The maximum RF Output Power (e.i.r.p.): _____dBm
The maximum (corresponding) Duty Cycle: _____ %

Equipment with dynamic behaviour, that behaviour is described here. (e.g. the different combinations of duty cycle and corresponding power levels to be declared):

f) The worst case operational mode for each of the following tests:

RF Output Power: 3.55dBm;
Power Spectral Density N/A;
Duty cycle, Tx-Sequence, Tx-gap N/A;
Accumulated Transmit time, Frequency Occupation & Hopping Sequence (only for FHSS equipment)
0.260s, 3, 79;
Hopping Frequency Separation (only for FHSS equipment) 1003kHz;
Medium Utilisation N/A;
Adaptivity Not Applicable;
Receiver Blocking Compliant;
Nominal Channel Bandwidth 1.0 MHz;
Transmitter unwanted emissions in the OOB domain -48.48 dBm/MHz;
Transmitter unwanted emissions in the spurious domain -32.84 dBm;
Receiver spurious emissions -62.56 dBm;

g) The different transmit operating modes (tick all that apply):

☒ Operating mode 1: Single Antenna Equipment
☒ Equipment with only 1 antenna
☐ Equipment with 2 diversity antennas but only one antenna active at any moment in time
☐ Smart Antenna Systems with 2 or more antennas, but operating in a (legacy) mode where only 1 antenna is used. (e.g. IEEE 802.11™ [i.3] legacy mode in smart antenna systems)

☐ Operating mode 2: Smart Antenna Systems - Multiple Antennas without beam forming
☐ Single spatial stream / Standard throughput / (e.g. IEEE 802.11™ [i.3] legacy mode)
☐ High Throughput (> 1 spatial stream) using Occupied Channel Bandwidth 1
☐ High Throughput (> 1 spatial stream) using Occupied Channel Bandwidth 2
Note: Add more lines if more channel bandwidths are supported.

☐ Operating mode 3: Smart Antenna Systems - Multiple Antennas with beam forming
☐ Single spatial stream / Standard throughput (e.g. IEEE 802.11™ [i.3] legacy mode)
☐ High Throughput (> 1 spatial stream) using Occupied Channel Bandwidth 1
☐ High Throughput (> 1 spatial stream) using Occupied Channel Bandwidth 2
Note: Add more lines if more channel bandwidths are supported.

h) In case of Smart Antenna Systems:

The number of Receive chains: ;
The number of Transmit chains: ;

☐ symmetrical power distribution
☐ asymmetrical power distribution

In case of beam forming, the maximum (additional) beam forming gain: N/A;

Note: The additional beam forming gain does not include the basic gain of a single antenna.

i) Operating Frequency Range(s) of the equipment:

Operating Frequency Range 1: 2402 MHz to 2480 MHz
Operating Frequency Range 2: _____ MHz to _____ MHz

Note: Add more lines if more Frequency Ranges are supported.

j) Nominal Channel Bandwidth(s):

Nominal Channel Bandwidth 1: 1.0 MHz
Nominal Channel Bandwidth 2: _____ MHz
Nominal Channel Bandwidth 3: _____ MHz
Nominal Channel Bandwidth 4: _____ MHz

Note: Add more lines if more channel bandwidths are supported.

k) Type of Equipment (stand-alone, combined, plug-in radio device, etc.):

☒ Stand-alone
☐ Combined Equipment (Equipment where the radio part is fully integrated within another type of equipment)
☐ Plug-in radio device (Equipment intended for a variety of host systems)
☐ Other _____;

l) The normal and extreme operating conditions that apply to the equipment:

Normal operating conditions (if applicable):

Operating temperature: +25 ° C

Other (please specify if applicable):

Extreme operating conditions:

Operating temperature range: Minimum: -40 ° C Maximum: +85 ° C

Other (please specify if applicable): …… Minimum: …… Maximum ……

Details provided are for the: ☒ Stand-alone equipment
☐ Combined (or host) equipment
☐ Test jig

m) The intended combination(s) of the radio equipment power settings and one or more antenna assemblies and their corresponding e.i.r.p levels:

Antenna Type:

☒ Integral Antenna (information to be provided in case of conducted measurements)

Antenna Gain: 1.39 dBi

If applicable, additional beamforming gain (excluding basic antenna gain):_____ dB

☐ Temporary RF connector provided
☐ No temporary RF connector provided
☐ Dedicated Antennas (equipment with antenna connector)
☐ Single power level with corresponding antenna(s)

☐ Multiple power settings and corresponding antenna(s)

Number of different Power Levels: .....

Power Level 1: ..... dBm
Power Level 2: ..... dBm
Power Level 3: ..... dBm

Note 1: Add more lines in case the equipment has more power levels.
Note 2: These power levels are conducted power levels (at antenna connector).

For each of the Power Levels, provide the intended antenna assemblies, their corresponding gains (G) and the resulting e.i.r.p. levels also taking into account the beamforming gain (Y) if applicable

Power Level 1: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 3: Add more rows in case more antenna assemblies are supported for this power level.

Power Level 2: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 4: Add more rows in case more antenna assemblies are supported for this power level.

Power Level 3: ..... dBm

Number of antenna assemblies provided for this power level: .....

Assembly #	Gain (dBi)	e.i.r.p.(dBm)	Part number or model name
1			
2			
3			
4			

NOTE 5: Add more rows in case more antenna assemblies are supported for this power level.

n) The nominal voltages of the stand-alone radio equipment or the nominal voltages of the combined (host) equipment or test jig in case of plug-in devices:

Details provided are for the: ☒ stand-alone equipment
☐ combined (or host) equipment
☐ test jig

Supply Voltage ☐ AC mains State AC voltage _____ V
☒ DC State DC voltage 3.3 V

In case of DC, indicate the type of power source
☐ Internal Power Supply
☐ External Power Supply or AC/DC adapter_____.
☐ Battery_____.
☒ Other: DC 3.3V .

o) Describe the test modes available which can facilitate testing:

Continuous transmiting&receiving and normal operation .

p) The equipment type (e.g. Bluetooth®, IEEE 802.11™ [i.3], IEEE 802.15.4™ [i.4], proprietary, etc.):

Bluetooth® .

q) If applicable, the statistical analysis referred to in clause 5.4.1 q)

(to be provided as separate attachment)

r) If applicable, the statistical analysis referred to in clause 5.4.1 r)

(to be provided as separate attachment)

s) Geo-location capability supported by the equipment:

☐ Yes
☐ The geographical location determined by the equipment as defined in clause 4.3.1.13.2 or clause 4.3.2.12.2 is not accessible to the user.

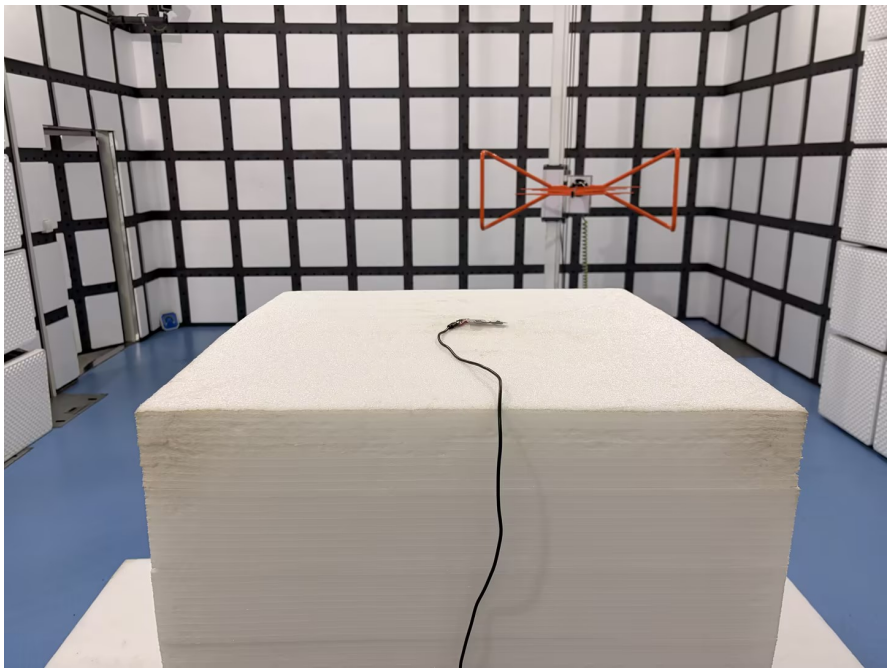
☒ No

EXHIBIT B - EUT PHOTOGRAPHS

Refer to Report No.: 2607T30160E-EM-02.

EXHIBIT C - TEST SETUP PHOTOGRAPHS

Radiated Spurious Emissions - Front View (Below 1GHz)



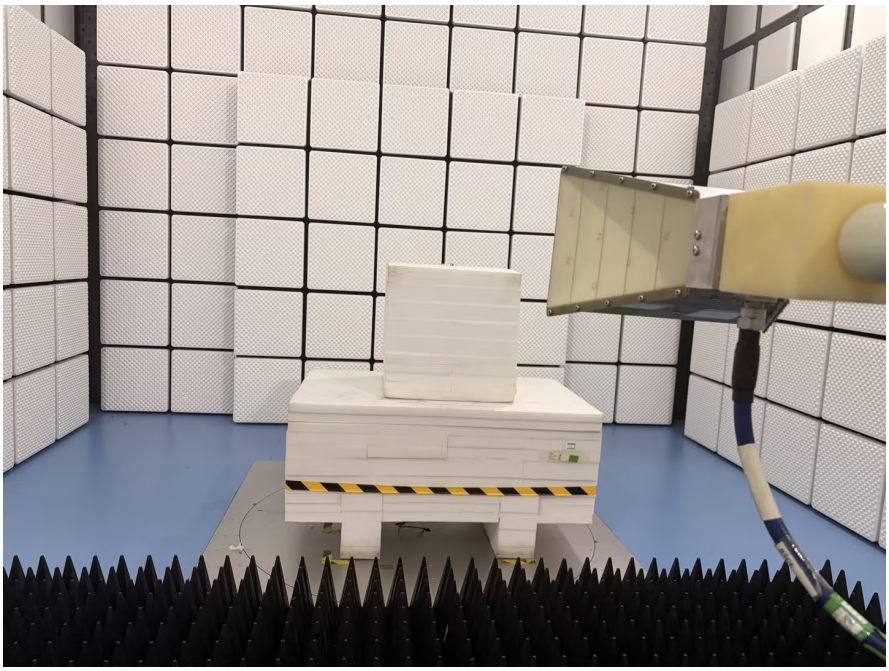
Radiated Spurious Emissions - Back View (Below 1GHz)



Radiated Spurious Emissions - Front View (Above 1GHz)



Radiated Spurious Emissions - Back View (Above 1GHz)



Declarations

1. Bay Area Compliance Laboratories Corp. (Fujian) is not responsible for authenticity of any information provided by the applicant. Information from the applicant that may affect test results are marked with an asterisk "★".

2. Unless otherwise stated the results shown in this test report refer only to the sample(s) tested.

3. Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty.

4. The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor K with the 95% confidence interval.

5. This report cannot be reproduced except in full, without prior written approval of the Company.

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******* END OF REPORT *******